SECTION 16745

TELECOMMUNICATION SYSTEMS

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PART 1 GENERAL COMMUNICATIONS REQUIREMENTS

GENERAL

1.1 Work Included

A. Work Specifically Excluded from Project: 1).

Incoming common carrier services. 2).

Private Branch Exchange Systems. 3).

Wide Area Network Systems.

4). Local Area Network Systems.

B. The Subcontractor shall be responsible for:

1). Providing additional materials, and the necessary labor and services required and ensure components of the system are completely installed in accordance with the intent of the Contract Documents.

2). Furnishing and installing incidental items not actually shown or specified, but which are required by good practice to provide complete functional systems.

3). Coordinating the details of facility equipment and construction for specification divisions, which affect the work covered under this Division.

4). Coordinating activities with the overall construction schedule.

5). Developing bill of materials, perform material management and efficient use of the materials whether they are issued by the Architect or purchased by the Contractor.

6). Ensure materials in excess of those required to complete the project are kept in their original condition and packaging for restocking.

7). Visit to site is required of all bidders prior to submission of bid. All contractors shall be responsible to familiarize themselves with all conditions that may affect their scope of work. No extra payment shall be allowed for work required because of these conditions, whether or not it is specifically mentioned in the project documents.

8). Intent of Drawings:

a). Unless specifically dimensioned, Communications plan drawings show only nominal locations of equipment, devices, raceways, cable trays, boxes, etc.
b). The Contractor shall be responsible for the proper placement and routing of equipment, cable, raceways, cable tray, and related components; according to the Contract Documents and subject to prior review by Contractor.
c). Refer conflicts between Contract Documents to Contractor, A/E, and Owner for resolution.

1.2 RELATED WORK

A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total general requirements for the project communications systems and equipment:

1). Division 1 sections included in the project specifications. 2).

The Subcontract.

1.3 REFERECES

A. Standards:

1). EIA/TIA-568-B 2001, Commercial Building Telecommunications Wiring Standard, or most current revision, shall be used.

2). EIA/TIA-569A, Commercial Building Standard for Pathways and Spaces, or most current revision, shall be used.

3). ANSI/TIA/EIA- 607 Commercial Building Grounding and Bonding Requirements for Telecommunications most current revision, shall be used.

4). International Standard ANSI/IEEE 802.1, 802.2, 802.3, 802.4 and 802.5. 5).

Standards related to or referenced by the standards listed herein.

6). National Electric Code (NEC) most current revision shall be used. 7).

NEMA.

8). Underwriters Laboratories (UL) 1863.

1.4 DEFINITIONS

A. Above Grade: Not buried in ground and not embedded in concrete slab on grade.

B. Active Equipment: Electronic equipment used to develop various WAN and LAN services, e.g., digital multiplexers RS 232 controllers, Ethernet hubs, token ring hubs, paging, etc.

C. Backbone: Collective term sometimes used to describe the campus and vertical distribution subsystem facilities and media interconnecting service entrances, communications rooms, and communications cabinets in all buildings.

D. Basket Tray: Continuous rigid welded steel, wire mesh cable management system, with minimum 4 inch sidewall height unless noted otherwise.

E. Below Grade: Buried in ground or embedded in concrete slab on grade of the building.

F. Bonding: Permanent joining of metallic parts to form an electrically conductive path which shall assure electrical continuity and the capacity to conduct safely any current likely to be imposed on it.

G. Building Communications Room (BCR): Room in each building used to distribute communications services to Satellite Communications Room/Satellite Communications Cabinets in the same building and interconnect with the Campus Communications Room. Typically, the Building Communications Room contains passive equipment used for electrical protection (protectors) and building cross connect, and active network equipment used for LANs. The Building Communications Room may also provide space to serve the functions of a Satellite Communications Room.

H. Cabinet: Freestanding, floor-mounted modular enclosure, designed to house and protect rack-mounted electronic equipment.

I. Cable Tray: Vertical or horizontal open supports, usually made of aluminum or steel, manufactured for the purpose of routing, supporting, and protecting communication cables. Minimum 6 inch solid side wall height unless noted otherwise.

J. Campus: Grounds and buildings of a multi-building premises environment.

K. Campus Communications Room (CCR): The room used to distribute communications services to building communications rooms on the premises, and to interconnect premises services with the public common carrier networks. Typically, the Campus Communications Room contains passive equipment used for electrical protection (protectors) and main campus cross connect, and active equipment used for PBX, WAN, and LAN.

L. Channel: The end-to-end transmission path between two points at which application-specific equipment is connected; may include one or more links, cross-connect jumper and/or patch cords, and work area station cords. Does not include connection to active equipment.

M. Concealed: Inside building above grade and located within walls, furred spaces, crawl spaces, attics, above suspended ceilings, in concrete. In general, any item not visible or directly accessible.

N. Connect: Complete hookup of item with required services, including conduit, cables, and other accessories.

O. Contractor: The company that is responsible for the over completion of the construction project. Also known as the General Contractor, or Construction Management Company.

P. Cross-Connect (field-voice): Equipment used to terminate and tie together communications circuits.

Q. Cross-Connect Jumper: A cluster of twisted-pair conductors without connectors used to establish a circuit by linking two termination points.

R. Exposed: Either visible or subject to mechanical or weather damage, indoors or outdoors, including areas such as mechanical and storage rooms. In general any item that is directly accessible without removing panels, walls, ceilings or other parts of structure.

S. Fiber Optic Distribution Unit (FDU): Cabinet with terminating equipment used to develop fiber optic cross-connect facilities.

T. Furnish: Supply and deliver complete.

U. Grounding: A conducting connection to earth, or to some conducting body that serves in place of earth.

V. Hinged Cover Enclosure: Wall-mounted box with a hinged cover that is used to house and protect electrical devices.

W. Horizontal: Pathway facilities and media connecting communications rooms to communication outlets.

X. Intermediate Distribution Frame: Additional horizontal cross-connect facility placed between the satellite communication room and communication outlet. The Intermediate Distribution Frame is used to distribute services to local areas. Typically, Intermediate Distribution Frames are hinged-cover containing passive equipment used for horizontal sub-system cross-connects.

Y. Jack: Receptacle used in conjunction with a plug to make electrical contact between communications circuits, e.g., eight-position/ eight-contact modular jacks.

Z. Link: A transmission path between two points, not including active equipment, work area cables, and equipment cables; one continuous section of conductors or fiber, including the connecting hardware at each end.

AA. Local Area Network (LAN): Data transmission facility connecting a number of communicating devices, e.g., serial data, Ethernet, token ring, etc. Typically, the network is limited to a single site.

BB. Media: Twisted-pair, coaxial, and fiber optic cable or cables used to provide signal transmission paths.

CC. Mounting Frame: A metal structure used to support wiring blocks, patch panels, and other communications equipment. Mounting frame that can be equipment racks or wall mounted.

DD. Multi-pair: Any common sheathed copper cable of 5 or more pairs.

EE. Owner: Micron Information Service (IS) Network Facilitization group is the systems owner under Division 17.

FF. Passive Equipment: Non-powered hardware and apparatus, e.g., equipment racks, cable trays, electrical protection, wiring blocks, fiber optic distribution unit, etc.

GG. Patch Cords: A cable assembly containing a length of copper or fiber cable with connectors on one or both ends. Patch cords are used to join communications circuits at a patch field.

HH. Patch Field (data): An assembly of passive equipment that provides the means to connect active equipment to backbone or horizontal cables. Patch cords are used to join communications circuits at a patch field.

II. Patch Panel: An assembly of terminal blocks and connectors used to provide the interface between fixed cable sub-system and patch cords.

JJ. Pathway: Facility for the placement of communications cable. A pathway facility can be composed of several components including conduit, wireway, cable tray, surface raceway, under-floor systems, raised floor, ceiling support wires, etc.

KK. Private Branch Exchange (PBX): Private communications switching system located on the user's premises. A PBX switches voice and data calls within a user's and between the premises and facilities provided by public common carrier networks.

LL. Protectors: Electrical protection devices used to limit foreign voltages on metallic communications circuits. Typically associated with metallic media exiting building to exterior.

MM. Provide: Furnish and install as defined above; perform work.

NN. Raceway: An enclosed channel designed expressly for holding wires or cables; may be of metal or insulating material. The term includes conduit, tubing, wire- ways, under-floor raceways, and surface raceways; cable tray is not defined as a raceway.

OO. Racks: An open, freestanding, floor-mounted structure, typically made of aluminum or steel, used to mount equipment; usually referred to as an equipment rack.

PP. Satellite Communications Cabinet (SCC): Cabinet or group of cabinets that may serve the same purpose as a Satellite Communications Room.

QQ. Satellite Communications Room (SCR): The Satellite Communications Room distributes communications services to users within a serving zone and interconnects with the Building Communications Room. Typically, the Satellite Communications Room contains passive equipment used for cross connect and active network equipment used for LANs.

RR. Contractor: The company that is responsible for the supply and installation of the Telecommunications systems under Division 17.

SS. Telecommunication Outlet (T.O.): Connecting device mounted in a work area used to terminate horizontal cable and interconnect cabling with station equipment.

TT. Tray Sidewall: Outside height dimension of tray. UU.

Underground: Buried in ground.

VV. Use (verb): Furnish and install as defined above.

WW. Wide Area Network (WAN): Active communications transmission facilities extending beyond the premises.

XX. Wiring Block: An assembly of punch down terminal blocks used to provide the interface between fixed cable sub-system and cross-connect jumper. Wiring blocks are used in cross-connect fields.

YY. Wireway: A supported pathway for cables, gutter.

1.5 SYSTEM DESCRIPTION

A. Communications shall implement a comprehensive integrated communications distribution system, as described in paragraph B below, to provide infrastructure which may be used to support one or more of the following services and systems:

1). Voice services.

2). Local area networks.

3). Data telecommunications.

4). Video services.

5). Wireless systems.

6). Facilities management systems.

B. The communications distribution system consists of the following seven major subsystems, as specified elsewhere.

1). Outside Plant Distribution Pathways 2).

Building Distribution Pathways

3). Communications Room Distribution Subsystem 4).

Campus Distribution Subsystem

5). Vertical Distribution Subsystem 6).

Horizontal Distribution Subsystem 7).

Work Area Distribution Subsystem

C. The communications distribution system is based on a combination of the following communications transmission technologies:

1). 100 ohm 4-pair Category 6A unshielded twisted-pair cable. 2).

50/125-micron multi-mode fiber optic cable.

3). 9/125 micron single mode fiber optic cable.

4). Category 6A compliant 8-position telecommunications jacks. 5).

Category 6A compliant patch cables.

6). Insulation displacement connector (IDC) type field terminated wiring blocks.

7). 50/125-micron multi-mode fiber optic patch cords 8).

9/125 micron single mode fiber optic patch cords.

1.6 DESIGN CRITERIA

A. Compliance by the Contractor with the provisions of this Specification does not relieve him of the responsibilities of furnishing materials and equipment of proper design, mechanically and electrically suited to meet operating guarantees at the specified service conditions.

B. The following are incorporated into the design:

1). The communications room installations are designed to be as similar as possible.

2). Electrical protection is provided for copper cabling, that has potential contact with lightning.

3). The location of communications rooms is intended to restrict the maximum horizontal subsystem wiring length (defined as a channel between a SCR/SCC cross-connect termination field and a served communication outlet to 90 meters (295 feet).

1.7 WARRANTY

A. Provide a single written Guarantee stating that all portions of the work are in accordance with Contract requirements, and shall satisfy all requirements to

qualify for the Leviton Network Solutions Lifetime System Warranty. Guarantee that all work shall interface and operate properly with telecommunications equipment provided under this section or other specification divisions.
Guarantee all work against faulty and improper material and workmanship for a MINIMUM period of one (1) year from date of final written acceptance by Owner, except that where manufacturers guarantee or warranty for longer terms, such longer term apply. Longer terms refer specifically to manufacturer warranties, especially those contingent upon work performance by a manufacturer Certified contractor. Within 24 hours after notification, correct any deficiencies, which occur during the guarantee period at no additional cost to Owner, all to the satisfaction of the Owner. Obtain similar guarantees from contractors, manufacturers, suppliers and sub trade specialists. Guarantee shall be submitted prior to contract close out.

B. For the NETWORK SYSTEM and as part of this warranty, provide a Leviton Network Solutions Lifetime System Warranty that shall:

1). Be free from product defects in materials and workmanship. 2).

Exceed applicable TIA/EIA and ISO/IEC Category 6A (where applicable), specifications in force at the time of installation.

3). Comply with design and performance requirements for recognized cabling media installed in the structured cabling link/channel.

4). Support any current or future application, which is designed for transmission over a structured cabling system as defined by the above referenced standard.

5). Be free from defects in material, workmanship and installation labor provided or carried out by the certified installer.

C. For all other systems, Guarantee all work against faulty and improper material and workmanship for a MINIMUM period of one (1) year from date of final written acceptance by Owner, except that where manufactures guarantee or warranty for longer terms, such longer term apply. Longer terms refer specifically to vendor or manufacturer warranties, especially those contingent upon work performance by a vendor's or manufacturer's Certified contractor.

D. In addition to the standard warranty requirements, the Contractor shall provide the following during the warranty period:

1). Within 24 hours after notification of a defect by the Owner or A/E, the Contractor shall start to make the necessary corrections and inform the Owner or A/E, as appropriate, of the planned corrective actions. The

Contractor shall follow this initial contact with continuous effort and complete any required corrective work within 15 days after notification.

2). A/E will retest a statistical sampling of horizontal cable, within one year (pending certified EIA/TIA test specifications), for conformance to standard transmission performance requirements. If any horizontal channels are found to be out of compliance, Contractor shall retest and recertify horizontal channels at no additional cost.

1.8 QUALIFICATIONS

A. Communications Pathway Installation: The Contractor shall have 5 years of documented experience installing raceway and cable tray systems for each of the types and system material components specified in the Contract Documents. In the case of newer technologies that do not have a 5-year history, the Contractor shall have documented experience for at least half of the lifetime of the new technology.

B. Communications Cabling: The Contractor shall have 5 years of documented experience performing cable placement, splicing, termination, connecting, and testing for each of the media types and system material components specified in the Subcontract Documents. In the case of newer technologies that do not have a 5-year history, the Contractor shall have documented experience for at least half of the lifetime of the new technology.

1). Project Manager's, A/E's, RCDD's, Supervisors, and Principal Skilled Technicians: Minimum of 5 years' experience in like work.

2). Copper and Optical Fiber installation technicians: Documented manufacturer training, licensing, and/or certification for the types of media specified, as applicable.

3). Contractor's Project Manager or Lead Technician and a minimum of 30% of the copper installation technicians must have successfully completed the Leviton certification training course.

1.9 REGULATORY REQUIREMENTS

A. Conformance to the latest national, state, and local codes and other legal requirements are the responsibility of the Contractor.

B. Contractor shall obtain permits and arrange inspections required by codes applicable to this Division and shall submit written evidence to Owner that the required permits, inspections, and code requirements have been secured.

Coordinate site permits with Owner and obtain Owner approval before contacting local governmental agencies.

C. Twisted-pair communications cables shall conform to National Electrical Code Article 800, Communication Circuits.

D. Fiber optic communications cables shall conform to National Electrical Code Article 770, Optical Fiber Cables. Cables installed within buildings shall be marked, by the manufacturer, with the appropriate certification code.

E. Contractor shall be responsible for insuring all materials being provided are new and unused and bear the Underwriters' Laboratory, Inc. label. If the UL label is not available, the Contractor must receive approval to use the specific product from the Owner.

F. Contractor shall be responsible for insuring all materials are clean, free of defects and corrosion and complies with the drawings and specifications as set forth.

1.10 SUBMITTALS

A. General:

1). Provide ongoing inspection and permit certificates and certificates of final inspection and acceptance from the authority having jurisdiction.

2). Manufacturer's standardized schematic diagrams and catalog cuts shall not be acceptable unless applicable portions of same are clearly indicated and non-applicable portions clearly deleted or crossed out.

3). When the specifications include product descriptions, model numbers, part numbers, etc. that have been superseded, changed, or discontinued, the Contractor shall submit a comparable substitution for review by the A/E. Final approval by Owner.

4). For deviations in product from these specifications, the Contractor shall submit a physical sample of the proposed product for review by the A/E and Owner or arrange for the A/E and Owner offsite visit to a sample product location when a product is too large for submittal.

B. Provide applicable portions of the following information for the attached sections in addition to the standard requirements with the Bid:

1). Documentation establishing qualifications to perform installation functions as required in the contract.

a). Contractor shall be fully certified as a Leviton Premier Network Installer (PNI).
b). Contractor shall be fully certified as a Corning Network Preferred Installer (NPI).
c). Contractor shall be approved by Owner.
d). Contractor must have a BICSI Registered Communications Distribution Designer (RCDD) on staff to oversee the project.
e). List of similar projects, of comparable complexity, type, and scope as detailed in the Contract Documents, and successfully completed by the bidder within the last 3 years.
f). Provide at least three references associated with these similar projects that the A/E, Owner or Contractor may contact and/or visit. For each, provide a description of the work performed; duration; and customer contact name, title, and phone number.
g). Summary of experience installing pathways, or copper and fiber optic communications systems as appropriate.
h). Resumes and backgrounds for key personnel that shall be involved in this specific project including those responsible for field supervision, quality assurance, and safety. Resumes shall include professional histories, references, training, licenses, and any certifications held. Any subsequent changes from these personnel must be agreed upon between Owner and Contractor.
i). Evidence of current and active certification as a Leviton Premier Network Installer and Corning Network Preferred Installer.

2). Statement demonstrating an understanding of project scope and schedule which includes the following information:

a). Where (city, office) the project shall be staffed. b). Project organizational chart with team names.

3). Provide the following information for materials, components, and equipment to be furnished by the Contractor:

a). Descriptive literature, manufacturer's specification data sheets, and manuals.
b). Individual price and delivery schedules. c). Manufacturer's product test data.
d). Final Performance testing criteria and data for communications distribution system cabling systems.

4). All Contractors shall be pre-qualified by Owner. Non pre-qualified contractors bids will not be accepted.

C. Provide applicable portions of the following information for sections listed in contract, in addition to the contract requirements within 7 days after award of Subcontract:

1). Project schedule in hard copy and electronic format compatible with Microsoft Project:

a). Include, at a minimum, major tasks, milestones, dependencies, staffing, and duration.
b). Contractor shall then work with General Contractor to merge this schedule into the overall construction schedule.

2). Provide a list of fiber optic and twisted-pair cable test equipment with calibration certification.

D. Provide applicable portions of the following information in addition to the contract requirements within 7 days after award of Subcontract:

1). Cable termination and test plans, in accordance with the "Communications Cable Testing Methods" section of this document.

2). Sketch showing large equipment containers and cable reel shipping sections, external dimensions, estimated weights, and rigging requirements.

E. Provide applicable portions of the following information in addition to the contract requirements, within 3 days after receiving each reel and/or box of cable:

1). Manufacturer's product test data for fiber optic cable components.

2). Proposed on-reel fiber optic testing sequence, expected duration, and scheduled date for testing.

3). Visually inspect Category 6A material. Material cannot be used before results have been submitted to Contractor and approved by A/E and Owner.

F. Provide applicable portions of the following completed test documentation in addition to the standard requirements, within 10 days after completion of the tests. Results to be submitted to Contractor and approval by A/E:

1). Test reports shall be submitted in accordance with the "Communications Cable Testing Methods" section of this document.

2). Submit test report documentation through the Contractor for review by the Owner and A/E for specification conformance.

3). Field test results for each cable reel with fiber optic components.

1.11 DEFINITION OF ACCEPTANCE

A. System acceptance shall be defined as that point in time when the following requirements have been fulfilled:

1). All submittals and documentation have been submitted, reviewed, and approved.

2). The complete system has successfully completed testing requirements. 3).

All Owners' staff personnel training programs have been completed.

4). All punch list items have been corrected and accepted.

1.12 PROJECT RECORD DOCUMENTS

A. Provide detailed project record documentation on paper and electronic format in addition to the standard requirements, within 14 days after completion of the work.

1). Maintain separate sets of redlined record drawings for the communications work that show the exact placement and identification of as-built system components.

2). Provide communication pathway record drawings which indicate exact placement and routing for components, e.g., maintenance holes, handholes, conduit, wireway, cable tray, pull boxes, enclosures, telecommunications outlet boxes, etc.

3). Provide communication room record drawings which indicate exact placement for components; e.g., conduit, wireway, cable tray, backboards, equipment cabinets, equipment racks, cross-connect equipment, etc.

4). Provide communication wiring and cabling record drawings and schedules which indicate exact placement, routing, and connection details for components, e.g., twisted-pair and fiber optic cables, splices, cable cross-connect termination locations, enclosures, telecommunications outlets, cross-connect jumpers, patch cords, etc.

5). Provide network schematics when appropriate.

B. Provide applicable portions of the following completed test documentation in addition to the standard requirements, within 10 days after completion of the tests. Results to be submitted to Contractor and approval by A/E:

1). Test reports shall be submitted in accordance with the "Communications Cable Testing Methods" section of this document.

2). Submit test report documentation through the Contractor for review by the Owner and A/E for specification conformance.

3). Twisted-pair cable field test documentation. 4).

Fiber optic cable field test documentation. 5).

Voice circuit field test documentation

PRODUCTS

2.1 GENERAL

A. Product Consistency: Each component shall be of 1 manufacturer. Where more than 1 manufacturer is specified, choose 1 manufacturer. Each type component shall be of 1 manufacturer only.

B. Part numbers and product codes in these specifications are correct as of the time of writing. Manufacturers may, however, change part numbers and product codes on short notice. In cases where part numbers or product codes differ from technical specifications for a particular product, provide product meeting the minimum technical specifications of the products in the specifications. The contractor shall notify the Owner of any product code and/or part number changes on the material list submittal and request written approval for their use from the Owner.

C. Manufacturer's Representatives: CoreTek Reps

EXECUTION

3.1 WORK LIMITATIONS

A. Protect benchmarks, existing structures, fences, utilities, sidewalks, paving, and curbs from equipment and vehicular traffic.

3.2 WORKMANSHIP

A. Manufactured products, materials, equipment, and components shall be provided, conditioned, applied, installed, connected, and tested in accordance with the manufacturer's specifications and printed instructions.

B. The installation of system components shall be carried out under the direction of qualified personnel. Appearance shall be considered as important as mechanical and electrical efficiency. Workmanship shall meet or exceed industry standards.

C. Place support for framing, raceways, cable trays, backboards, equipment racks, and cabinets under the provisions of the "Communications Supporting Methods" section of this document.

D. Components and equipment having a maximum weight, including contents, exceeding 400 pounds shall be installed in accordance with Section 05455 - Seismic Restraints Criteria.

3.3 SERVICE CONTINUITY

A. Maintain continuity of communications services to functioning portions of the process or buildings during hours of normal use.

B. Arrange temporary outages for cutover work with Owner. Keep outages to a minimum number and a minimum length of time in order to provide minimum impact.

3.4 BASELINE BENCHMARKS

A. Basic indoor and outdoor horizontal and vertical base lines, control points, and benchmarks, as appropriate, shall be established by Contractor. Use these references to determine offsets and locations for maintenance hole, hand-hole, duct bank, raceway, cable tray, pull box, equipment, and cable layout work under this Subcontract.

B. For locations where additional surveying is required to determine the location of the communications utility corridor, Contractor shall provide construction staking.

C. Provide requests for line and grade marking or work alignment verification to Contractor a reasonable time in advance of the need, to allow for coordination with the civil contractor. Include the times and places where this service is required.

3.5 LAYOUT AND TOLERANCES

A. Follow as closely as practicable the schematic design shown on the Drawings. Make necessary measurements in the field to verify exact locations and ensure precise location and fit of specified items in accordance with the Drawings. Make no substantial alterations without prior approval of Contractor, the A/E and Owner.

B. Perform work to the lines, grades, and elevations indicated on the drawings. Provide experienced and competent personnel to locate and lay out the work and provide them with suitable tools, equipment, and other materials required to complete layout and measurement work. Use lasers or other approved methods to establish line and grade.

C. Tolerances: After indoor and/or outdoor placement, align conduit, wireway, cable tray, boxes, and equipment. Following this, check for conformance to line and grade. The deviation of any pathway run or item of equipment from line and grade shall not exceed 1/2 inch, unless the specifications or Drawings indicate otherwise.

3.1 CONSTRUCTION REVIEW

A. The A/E, General Contractor and Owner will review and observe installation work to ensure compliance by the Contractor with requirements of the Contract Documents.

B. The Contractor shall inspect and test completed communications installations to demonstrate specified performance levels including the following:

1). Furnish instruments and personnel required for the inspections and tests.

2). Perform tests in the presence of the A/E, Contractor and Owner.

3). Allow 48 hours notice to Owner for all testing to allow observance of testing.

4). Demonstrate that the system components operate in accordance with the Contract Documents.

C. Review, observation, assistance, and actions by the A/E, Contractor and Owner will not be construed as undertaking supervisory control of the work or of methods and means employed by the Contractor. The A/E's and Contractor review and observation activities will not relieve the Contractor from the responsibilities of these Contract Documents.

D. The fact that the Contractor, A/E or Owner does not make early discovery of faulty or omitted work will not bar the Contractor, A/E or Owner from subsequently rejecting this work and insisting that the Contractor make the necessary corrections.

E. Regardless of when discovery and rejection are made, and regardless of when the Contractor is ordered to correct such work, the Contractor Shall have no claim against the A/E or Owner for an increase in the Subcontract price, or for any payment on account of increased cost, damage, or loss.

PART 2 COMMUNICATIONS GROUNDING

GENERAL

4.1 SUMMARY

A. Section Includes:

1). Communications ground conductors.

2). Equipment, raceway, cable tray, and cable ground connections. 3).

Ground busbar hardware.

4). Equipment rack and cabinet grounding. 5).

Splice case grounding.

6). Protectors.

7). Cable sheath grounding.

B. Related Specifications:

1). Division 1 specifications.

C. Standards:

1). ANSI/TIA/EIA-607 Commercial Building Grounding and Bonding Requirements for Telecommunications.

4.2 SYSTEM DESCRIPTION

A. Communications system grounding and bonding extends beyond the grounding electrode system and building ground ring provided for in Specification 16060 - Grounding. See Figure 1.

B. Ground and bond metallic communications conduit, wireway, and cable tray systems.

C. Provide building and telecommunications ground busbar hardware at Communications Rooms and connect to pigtail extensions off the building—grounding ring provided by the electrical subcontractor to make a Telecommunications Grounding and Bonding network.

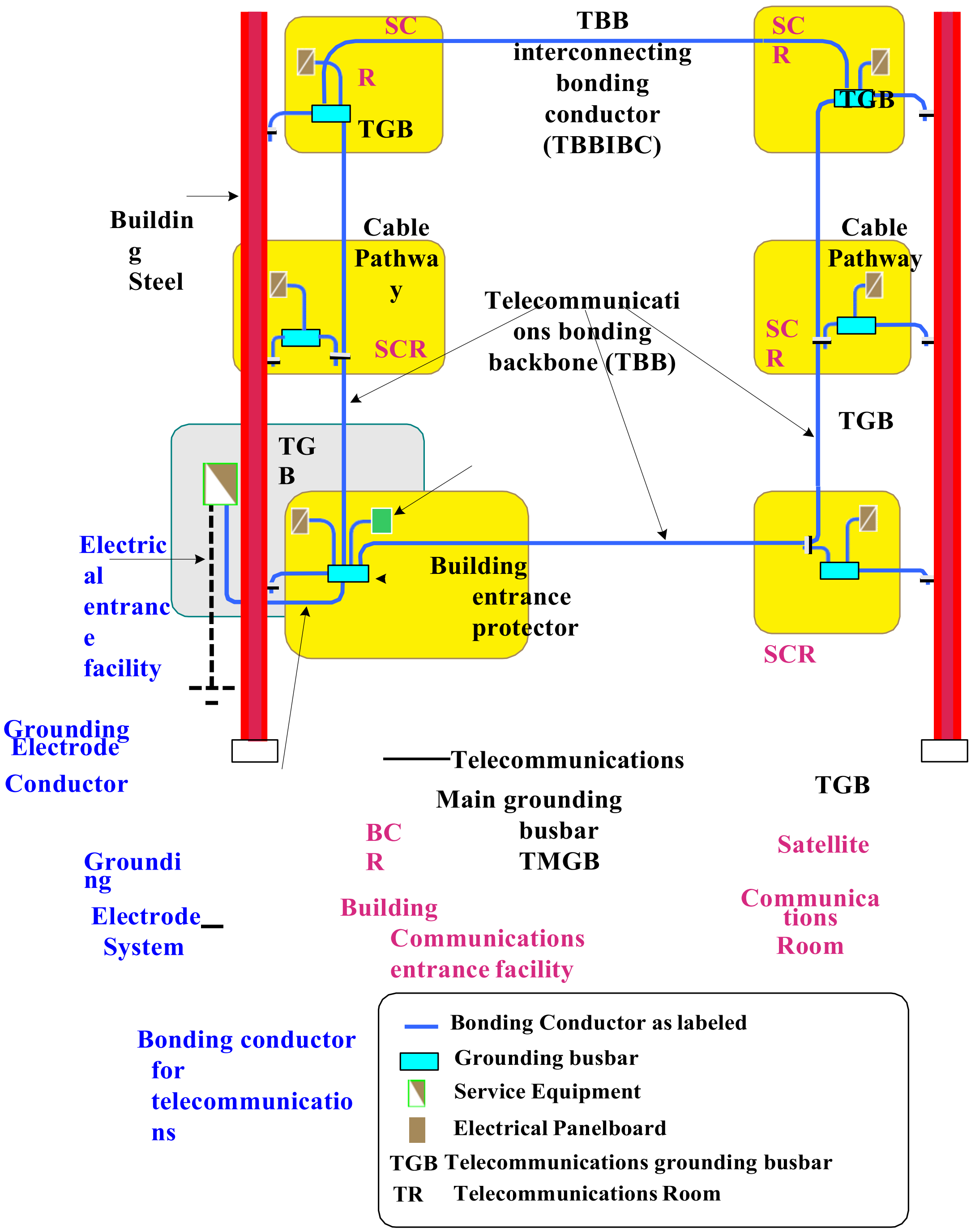
D. Bond ground bars and grounding cables on equipment cabinets and racks.

E. Bond together exposed non-current-carrying metal parts of communications equipment, metal raceway, and cable tray systems.

F. Ground and/or bond exposed metallic shields in multi-pair communications cables.

G. A Telecommunications Grounding and Bonding Network's basic function is to reduce or equalize potential differences between telecommunications systems bonded to it. A TGBN is not intended to serve as the only conductor providing a ground fault current return path.

Schematic of a Telecommunications Grounding and Bonding Network



This schematic example of a grounding and bonding network contains terminology according to ANSI/TIA/EIA-607

Figure 1

4.3 DEFINITIONS

A. Building Communications Room (BCR):

1). Is a centralized multipurpose room for telecommunications equipment that serves the occupants of a building and is considered distinct from a SCR because of the nature or complexity of the equipment,

2). Is also an entrance to a building for both public and private network service cables, joining inter- or intra-building telecommunications backbone facilities,

3). A space for housing telecommunications equipment, cable terminations, and cross-connect cabling that services one area or floor of a building. Is the location for the cross connects between the backbone and horizontal facilities,

4). It also may serve as a Computer Room housing a limited number of Servers.

B. Satellite Communications Room (SCR): An enclosed space for housing telecommunications equipment, cable terminations, and cross-connect cabling that services one area or floor of a building. The SCR is the location for the cross connects between the backbone and horizontal facilities.

C. Telecommunications Bonding Backbone (TBB): is a conductor extending from the telecommunications main grounding busbar to the farthest floor TGB that interconnects all TGBs with the TMGB in a logical ring.

D. Telecommunications Grounding Busbar (TGB): A busbar placed in the SCR in a convenient and accessible location and bonded by means of the bonding conductor for telecommunications equipment grounding.

E. Telecommunications Main Grounding Busbar (TMGB): A busbar placed in the BCR in a convenient and accessible location and bonded by means of the bonding conductor for telecommunications to the service equipment (power) ground.

4.1 SUBMITTALS

A. Submit in accordance with Specification 01300.

B. Pre-construction:

1). Submit a Testing Plan that defines the method to be used for testing the telecom grounding system and give sample data sheets.

C. Assurance/Control Systems:

1). Submit a Communication Room/ communication cabinet ground terminal block schedule listing test results which verify ground connection loss from each telecommunication room ground to the building main ground pigtail is 1 ohm or less.

2). Submit test results to verify that resistance from intra-room cable tray systems to telecommunications grounding busbars is less than 1 ohm. Measurements shall be taken from section of cable tray section furthest from telecommunications grounding busbars.

3). Submit test results for racks and cabinets to telecommunications ground busbar, located within Communication Rooms, which verify resistance is less than 1 ohm.

4). Submit test results to verify resistance of ground connections made from Computer Room raised floor to building ground busbars is less than 1 ohm.

5). Submit test results of multi-pair cable metallic shields to verify that resistance to telecommunications ground busbar is less than 1 ohm.

6). Submit test results to verify that resistance from communications backboards ground connector to telecommunication ground busbar is less than 1 ohm.

7). Submit test results to verify resistance of telecommunications equipment, located in maintenance holes, to ground busbar, also located in maintenance hole is less than 1 ohm.

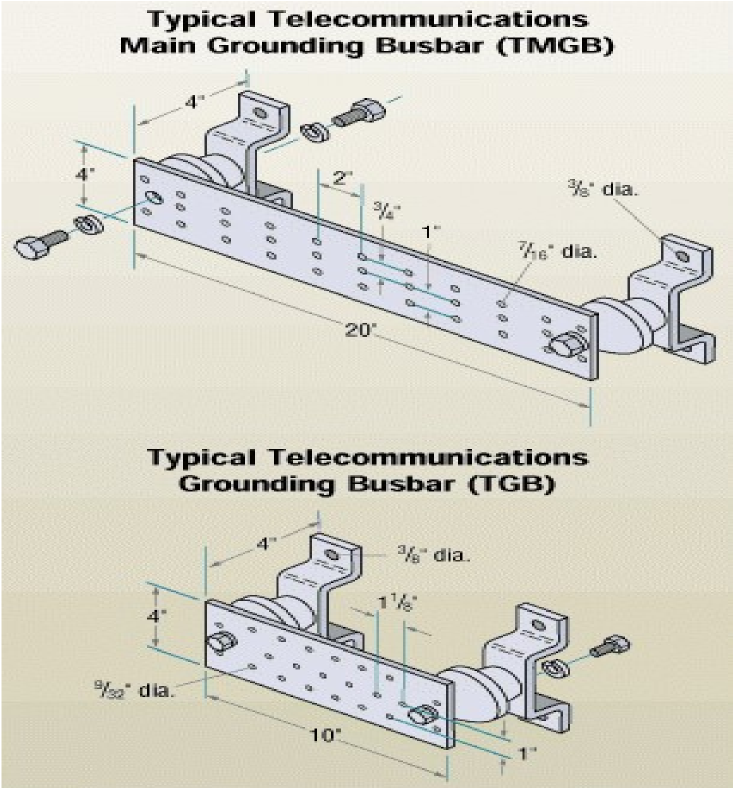
<u>PRODUCTS</u>

5.1 GROUND WIRE

A. Grounding conductors of the size and type shown on the Drawings, or use a minimum of green No. 6 AWG insulated copper wire.

5.2 GROUND BUSBAR

A. For the TMGB or TMB, provide a solid copper busbar, predrilled for two-hole lug connections, with a minimum thickness of 1/4 inch, for wall and backboard mounting using standoff insulators as shown below. See figure 2.



The general arrangements and dimensions of a typical telecommunications main grounding busbar (TMGB) and telecommunications grounding busbar (TGB).

B. Figure 2

5.3 GROUND CONNECTIONS

A. Below Grade: Exothermic-welded type connectors.

B. Above Grade:

1). Bonding Jumpers: Compression type connectors using zinc-plated fasteners and external tooth lock washers.

2). Ground Busbars: Two-hole compression type lugs using tin-plated copper or copper alloy bolts and nuts.

3). Rack and Cabinet Ground Bars: One-hole compression-type lugs using zinc-plated or copper alloy fasteners.

C. Cable Shields: Make ground connections to multi-pair metallic shielded communications cables using shield bonding connectors with screw stud connection by AT&T, 3M, or Marconi OSP.

5.4 CORROSION INHIBITORS

A. When making ground bond connections in areas where increased risk of corrosion is anticipated, apply a conductive corrosion inhibitor to contact surfaces. Use conductive corrosion inhibitor appropriate for protecting a connection between the metals used. Do not use corrosion inhibitor in clean room areas rated Class 100 or better.

5.5 EQUIPMENT CABINET GROUND BARS

A. Provide solid copper ground bars designed for mounting on the framework of open or cabinet-enclosed equipment racks with minimum dimensions of 1/4- inch thick by 3/4-inch wide. As indicated on project drawings.

5.6 FREE STANDING EQUIPMENT RACK GROUND CONNECTION

A. Provide ground jumper from mounting hole on rack to equipment ground location.

5.7 GROUND STRIP

A. Telecommunication ground strips used in maintenance holes/handholes shall provide coverage throughout the length of the maintenance hole/handholes. Width of 1 inch and a thickness of 1/8 inch.

5.8 GROUND JUMPER

A. Where painted mounting brackets are used or where there is no electrically continuous ground path between telecommunication racks and telecommunications equipment, provide a grounding connection between telecommunications equipment and telecommunications racks using green No. 6 AWG ground conductor.

5.9 TELECOMMUNICATION GROUND RODS

A. Telecommunications ground rods shall be used in maintenance hole applications and shall be constructed of copper or copper clad steel, 5/8 or 3/4 inch in diameter, 10 feet long, driven full length into the earth.

5.10 GROUND TERMINAL BLOCKS

A. At any equipment mounting location (e.g., backboards and hinged cover enclosures) where rack-type ground bars cannot be mounted, provide screw lug-type terminal blocks.

5.11 SPLICE CASE GROUND ACCESSORIES

A. Splice case grounding and bonding accessories shall be of the same manufacture as the splice case for grounding in maintenance holes. If the splice case manufacturer does not off a grounding kit, then as a minimum, a use green No. 6 AWG insulated ground wire and a shield bonding connector and bond the case to the ground strap in the maintenance hole.

5.12 TRAY

A. When grounding tests indicate greater than 1 ohm, bonding jumpers shall be installed at every joint.

B. Physically bond all metal conduits to the cable tray at locations where cable leaves the cable tray. If the metal conduit is not physically connected to the cable tray then provide a bonding jumper between the conduit and the cable tray.

EXECUTION

6.1 INSTALLATION

A. General Ground System Installation:

1). Approved building ground sources are specified by Specification 16060 - Grounding.

2). Furnish and install wire and hardware required to properly ground, bond, and connect communications raceway, cable tray, metallic cable shields, and equipment to a ground source.

3). Bonding jumpers shall be continuous with no splices. Use the shortest length of bonding jumper possible.

4). Provide ground paths that are permanent and continuous with a resistance of 1 ohm or less from raceway, cable tray, and equipment connections to the building—grounding electrode. The resistance across individual bonding connections shall be 10 milliohms or less.

5). Below-Grade Grounding Connections: When making thermite welds, wire brush or file the point of contact to a bare metal surface. Use thermite welding cartridges and molds in accordance with the manufacturer's recommendations. After welds have been made and cooled, brush slag from the weld area and thoroughly clean the joint and area. Notify Contractor prior to backfilling any ground connections.

6). Above-Grade Grounding Connections: When making bolted or screwed connections to attach bonding jumpers, remove paint to expose the entire contact surface by grinding where necessary; thoroughly clean connector, plate, and other contact surfaces; and apply an appropriate corrosion inhibitor to surfaces before joining.

7). Bonding Jumpers:

a). Use green insulated ground wire of the size and type shown on the Drawings or use a minimum of green No. 6 AWG insulated copper wire.

8). Assemble bonding jumpers using insulated ground wire terminated with compression connectors.

9). Use compression connectors of proper size for conductors specified. Use connector manufacturer's compression tool.

B. Bonding Jumper Fasteners:

1). Conduit: Fasten bonding jumpers using screw lugs on grounding bushings or conduit strut clamps, or the clamp pads on push-type conduit fasteners. When screw lug connection to a conduit strut clamp is not possible, fasten the plain end of a bonding jumper wire by slipping this plain end under the conduit strut clamp pad; tighten the clamp screw firmly. Where appropriate, use zinc-plated external tooth lock washers.

2). Wireway and Cable Tray: Fasten bonding jumpers using zinc-plated bolts, external tooth lock washers, and nuts. Install protective cover, e.g., zinc-plated acorn nuts, on any bolts extending into wireway or cable tray to prevent cable damage.

3). Ground Plates and Busbars: Fasten bonding jumpers using two-hole compression lugs. Use tin-plated copper or copper alloy bolts, external tooth lock washers, and nuts.

4). Steel Strut and Raised Floor Stringers: Fasten bonding jumpers using zinc-plated, self-drill screws and external tooth lock washers.

C. Communication Room Grounding:

1). Building Ground Busbars: Provide ground busbar hardware at each Communications Room and connect to pigtail extensions of the building—grounding ring provided by others. Verify that the ground ring pigtail is the same type and size conductor used for the main building ground ring. This ground busbar is not required if a building ground busbar is available in the room.

2). Telecommunications Ground Busbars:

a). Provide Communications Room telecommunications ground busbar hardware at cable tray height as indicated on the Drawings.
b). Connect the telecommunications ground busbar to the building ground busbar located in the same room using two-hole compression lugs and a grounding jumper of the same size as the pigtail extension of the main building grounding ring.

3). Cable Tray Systems: Make ground connections by installing the following bonding jumpers:

a). Install a green No. 6 AWG bonding jumper between the telecommunications ground busbar and the nearest access to the cable tray.
b). Use green No. 6 AWG bonding jumpers across cable tray junctions

4). Self-Supporting and Cabinet-Mounted Painted Equipment Rack Ground Bars:

a). If racks are painted so as not electrically continuous, daisy chain bond each rack together and connect closest rack to room bus bar using a green No. 6 AWG bonding jumper.

5). Backboards: Provide a screw lug-type terminal block or drilled and tapped copper strip near the top of backboards used for communications cross-connect systems. Connect backboard ground terminals to the aluminum pan in the telephone-type cable tray using an insulated green No. 6 AWG bonding jumper.

D. Other Communication Room Ground Systems: Ground metallic conduit, wire- ways, and other metallic equipment located away from equipment racks or cabinets to the cable tray or the telecommunications ground busbar, whichever is closer, using insulated green No. 6 AWG ground wire bonding jumpers.

E. Communications Cable Grounding:

1). Bond metallic cable sheaths in multi-pair communications cables together at each splicing and/or terminating location to provide 100 percent metallic sheath continuity throughout the communications distribution system.

a). At termination points, install a cable shield bonding connector to provide a screw stud connection for ground wire. Use a bonding jumper to connect the cable shield connector to an appropriate ground source like the communications room rack or cabinet ground bar.

b). Bond metallic cable shields together within splice closures using cable shield bonding connectors or the splice case grounding and bonding accessories provided by the splice case manufacturer.
When an external ground connection is provided as part of splice closure, connect to an approved ground source and other metallic components and equipment at that location; i.e., Maintenance Hole Grounding System.

F. Manufacturing Area Telecommunications Grounding:

1). Conduit: Ground and bond metallic conduit systems as follows: a).

Ground metallic conduit at each end using green No. 6 AWG bonding jumpers.

b). Bond at intermediate metallic enclosures and across joints using green No. 6 AWG bonding jumpers.

c). Bond every subfab level metallic conduit penetrating the fab floor slab at the top steel strut rack support as follows:

1. Unpainted Steel Strut: When conduit/strut clamps are used, no additional bonding is required.
2. Painted Steel Strut: Use green No. 6 AWG bonding jumpers between steel strut structure and conduit. Connect to conduit

by slipping plain end of bonding jumper under conduit/strut clamp pad.

d). Bond every metallic conduit which penetrates the plane of the fab level raised floor to the raised floor stringer system as follows:
1. Unpainted Steel Strut Attached to Raised Floor System: When conduit/strut clamps are used to attach conduit to steel strut, no additional bonding is required.
2. Unpainted Steel Strut Which is not attached to Raised Floor System: When conduit/strut clamps are used to attach conduit to steel strut, use green No. 6 AWG bonding jumpers to connect steel strut to the raised floor system.
3. Painted Steel Strut: Use green No. 6 AWG bonding jumpers to connect conduit to the stringer system. Mount a push-type conduit fastener onto every metallic conduit. Place fasteners no higher or lower than 3 inches from the raised floor stringer.

2). Wireway (gutter/wiremold): Ground and bond metallic wireway systems as follows:

a). Bond the metallic structures of wireway to provide 100 percent electrical continuity throughout the wireway system by connecting green No. 6 AWG bonding jumper at intermediate metallic enclosures and across section junctions.
b). Install insulated green No. 6 AWG bonding jumpers between the wireway system bonded as required in paragraph 1 above, and the closest building ground at each end and approximately every 48 feet.
c). Bond every metallic wireway penetrating the floor slab at the top steel strut rack support using green No. 6 AWG bonding jumpers.
d). Bond every metallic wireway that penetrates the plane of the raised floor to the raised floor system using green No. 6 AWG bonding jumpers. Locate wireway connection no higher or lower than 3 inches from the raised floor stringer.

3). Cable Tray Systems:

a). Bond the metallic structures of one cable tray in each single or multiple tray run following the same path to provide 100 percent electrical continuity throughout this cable tray system as follows:
1. Splice plates provided by the cable tray manufacturer can be used for providing a ground bonding connection between cable tray sections when the resistance across a bolted connection is 10 milliohms or less. The subcontractor shall verify this loss by testing across one slice plate connection.
2. Install a green No. 6 AWG bonding jumper across each cable tray splice or junction where splice plates cannot be used.

b). Install insulated green No. 4/0 AWG bonding jumpers between the cable tray bonded as required in paragraph C.1 above, and column-mounted building ground plates (pads) approximately every 48 feet.

c). Bond the metallic structures of the cable tray bonded in paragraph 1 above to the following systems:

1. Provide green No. 6 AWG bonding jumpers between the bonded cable tray and the main steel strut support structure approximately every 48 feet.

G. Non-manufacturing Area Grounding:

1). Conduit: Use insulated green No. 6 AWG bonding jumpers to ground metallic conduit at each end and bond at intermediate metallic enclosures.

2). Wireway: Use insulated green No. 6 AWG bonding jumpers to ground or bond metallic wireway at each end at intermediate metallic enclosures and across section junctions.

3). Cable Tray Systems: Use insulated green No. 6 AWG bonding jumpers to ground cable tray to building ground plates at each end and approximately every 48 feet.

H. Maintenance hole Grounding:

1). Provide ground rod, copper or copper clad steel, 5/8 inch in diameter, 10 feet long, driven full length into the earth. Install ground rod outside maintenance holes in the conduit raceway trench. Connect ground rods to flat copper strip inside the maintenance hole with a green No. 6 AWG copper ground conductor.

2). Provide telecommunications flat copper strip on the interior wall surface of the maintenance hole which is bonded to grounding rod or building ground grid, whichever is used, and to pre-cast reinforcing bar structure.

a). Bonding surface shall be free and clean of paint.

b). Conductive Corrosion inhibitor shall be applied to bonding locations in areas expected to promote corrosion. Use oxidation-inhibitor appropriate for protecting the connection between metals used.

3). Ground and bond telecommunication metallic conduit and ducting to telecommunication flat copper strip using a green No. 6 AWG ground conductor. Length of conductor shall be kept as short as possible.

6.2 FIELD QUALITY CONTROL

A. Comply with requirements of Specification 01400 - Quality Control.

B. Site Tests:

1). Test resistance between ground connections using a micro-ohm meter: a).

Test provided Communication Room telecommunications ground busbars, enclosure ground terminal strips to the grounding electrode connection at the building main ground busbar. Verify loss is 1 ohm or less.

b). Test provided raceway and cable tray bonding connections located in manufacturing areas. Verify loss across bonding jumpers or bonding connections is 10 milliohms or less.

c). Submit final field test documentation in list form.

2). Inspect grounding and bonding system conductors and connections for tightness and proper installation.

PART 3 COMMUNICATIONS CABLE TESTING METHODS

GENERAL

7.1 SUMMARY

A. Section Includes:

1). Unshielded twisted-pair (UTP) cable. 2).

Fiber optic cable.

3). Composite horizontal cable. 4).

Multi-conductor connectors. 5).

Fiber optic connector.

6). Ground wire.

7). Shielded twisted pair (STP) cable.

B. References:

1). ANSI / TIA/EIA – 526-7

a). OFSTP-7 Measurement of Optical Power Loss of Installed Single- Mode Fiber Cable Plant.

2). ANSI / TIA/EIA – 526-14A

a). OFSTP-14 Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant.

3). ANSI / TIA/EIA 568A – B.1

a). Commercial Building Telecommunications Cabling Standard, Part 1: General Requirements Clause 1.1

7.2 SUBMITTALS

A. Provide applicable portions of the following information in addition to the contract requirements within 10 days after award of Subcontract:

1). Cable termination and test plans.

B. Include in the test plan submittal:

1). The test equipment shall be used, for Owner approval.

2). A sample of the printed test results for each type of testing.

C. Closeout Submittals:

1). One binder of the compiled individual test data sheets:

a). One set with each Horizontal UTP cable tested. b).
One set with each fiber optic cable strand tested.
c). One set with each backbone unshielded twisted pair (UTP) cable multi pair test log.

7.3 CABLE TERMINATION AND TEST PLANS

A. General:

1). Provide proof of testing technician(s) certification for operation of the specific units of test equipment that are proposed for use.

2). The Subcontractor shall obtain Contractor and Owner approval for each termination and test plan prior to execution of the work.

3). This Section covers work necessary to furnish communications system testing, including the following:

a). On-the-reel fiber optic cable testing, as required/requested. b).
Backbone unshielded twisted pair (UTP) cable system testing. c).
Horizontal UTP cable system testing.
d). Fiber optic cable system testing.

4). Inspection Requirements:

a). As part of any performance test, inspect cable, material, and equipment for physical damage, continuity, and proper connection.
b). Verify identification and labeling at required locations for visibility, condition, legibility, and accuracy.

5). Test Report Requirements: Each test report shall include the following sections:

a). Scope of testing.

b). List of equipment used in the test with a photocopy of the calibration certificate.
c). List of technicians performing the tests identified in the scope of testing.
d). Summary of test results: The summary shall be completed by the Contractor. The Test Summary shall utilize the available columns provided in the cable schedules. Electronic copies of the summary forms shall be made available to the Contractor, A/E and Owner upon request.
e). Individual test data sheets: The individual test data sheets shall be developed and completed by the subcontractor. Formatted output from cable scanners and OTDRs are typically acceptable provided they contain the information required by this Section.
f). The Owner will have the option of witnessing all testing. Provide 48 hours notice to Owner (MTI I.S. Network Facilitations), prior to testing.

B. Fiber Optic and Twisted-Pair Cable Termination Plans:

1). Submit detailed termination plans for both fiber optic and twisted pair cables that describe how each system component shall be installed and terminated.

2). In addition, provide descriptions of proposed methods for connectorizing and terminating the fiber optic cable.

C. Fiber Optic and Twisted-Pair Cable Test Plans:

1). Submit detailed test plans for both fiber optic and twisted-pair cable channels which include at least the following information:

a). Describe the tests to be performed.
b). Explain when and how each system component shall be tested. c).
List the test equipment to be used.
d). Itemize how theoretical loss budgets and test parameters shall be calculated and listed.
e). Provide an example of the test reporting documentation for each type of test that provides a written verification of the results as required in paragraph 2 below.

2). Provide testing documentation which includes: a).

Project name.
b). Name of subcontractor company. c).
Date document was prepared.

d). Dates and times of test.
e). Personnel performing tests. f).
Media type.
g). Make Model and serial number of test equipment used. h).
Initial test results.
i). Description of discrepancies found or failure, if any. j).
Corrective action, if any.
k). Date and person performing corrections. l).
Retest results, if required.
m). Include space for Contractor sign-off.
n). Copy of test equipment calibration certificates.

3). Fiber Optic Test Documents also include: a).

Cable tag number.
b). Cable bandwidth specification. c).
Fiber count.
d). Connector type.
e). Cable length per footage markers. f).
Measured cable length.
g). Calculated maximum link loss.
h). Measured link loss for each fiber from each end, measured at:
1. Multimode fiber: 850 and 1300 nanometers.
2. Single mode fiber: 1310 and 1550 nanometers.

4). Twisted-Pair (Four Pair) Data Cable Tests Documents also include: a).

Cable category and type.
b). Cable number.
c). Measured cable length.
d). Measured link losses with graphs.
e). Graphic or tabular data for all parameters tested per TIA/EIA procedures below.

5). Large Count Twisted-Pair Phone Cable Tests Documents also include: a).

Cable type.
b). Pair count.
c). Cable number.
d). Measured cable length.
e). Tabular data for all parameters tested for each pair. f).
Total number of serviceable pairs in cable.

PRODUCTS

8.1 ***NOT USED***

EXECUTION

9.1 GENERAL

A. Cables that fail tests shall be repaired or replaced at the discretion of the Owner, and re-tested without additional cost of the Owner. Do not abandon cables in place.

B. Perform testing after complete installation of the cable system. Piecemeal testing is not acceptable.

9.2 FIBER OPTIC CABLE TESTS

A. The common tools for testing are a power meter with a stabilized light source (power loss test set) and a recording optical time domain reflectometer (OTDR).

B. Fiber Optic Test Plans: Provide separate schemes for the following activities: 1).

On-reel cable testing performed prior to installation, as required/requested.

a). Inspect cables for signs of physical damage b).
OTDR Test:
1. At a minimum, an OTDR shall be used to: determine the exact length of the fiber, and to identify the presence of any core damage.

c). Method: Using a temporary connector, test the fiber on the real to detect internal defect that may have been caused in shipping.
Where defects are found inherent in the fiber itself: replace any cable having fewer than the manufacturers guaranteed number of serviceable fibers.

2). Interbuilding (Campus Subsystem) fiber optic segment post-installation test and Intrabuilding (Vertical and Horizontal Subsystem) fiber optic segment post-installation test:

a). Final Acceptance Power Loss Test:
1. The Contractor shall test all fiber links for overall attenuation from end to end. The Contractor shall perform attenuation using a stabilized light source and measure the received level with a power meter.
2. Method: The Contractor shall perform the following measurement attenuation tests using the insertion method. The Contractor shall measure the attenuation of the fiber optic network, which makes use of plug connectors and splices as

coupling points. Attenuation shall be measured between all the coupling points (when applicable) using the insertion method. The Contractor must first determine a reference measurement to determine the injection power level of the stabilized source. The Contractor shall connect the source directly to the optical power level meter using the referenced cable and connection. The reference level shall be checked and documented periodically in decibels or decibels per meter throughout the acceptance tests. The optical source shall then be connected to the beginning point in the network and the optical power level meter transferred to the remote end by a member of the test crew. The received level at this point shall be measured. The measured attenuation shall be obtained by subtracting the reference level from the received level.

3. Perform the End to end, bi directional attenuation (loss) acceptance test at the 850-nm and 1300-nm wavelength for multimode fiber, and the 1310-nm and 1550-nm wavelength for single mode fiber. Conduct tests in accordance with EIA/TIA 526 14, Method B and with test instrument manufacturers printed instructions.
4. When differences are found between the 2 tests on a single fiber use the OTDR to: determine the true loss for each connector pair, the exact length of the fiber to the damage point, and to identify the presence of any core damage.
5. The Contractor shall perform the attenuation acceptance test from end to end. The Contractor shall perform the tests utilizing a stabilized light source and measure the received level with a power meter. The Contractor shall document all test performed and submit results to the Owner on Contractor supplied (and Owner approved) forms.
6. Faults related to connectorization shall be corrected, and the fiber re tested as stated in paragraph 3.2.b and c above, until acceptable attenuation measurements are recorded at no expense to owner.
7. Where defects are found inherent in the fiber itself: replace any cable having fewer than the manufacturers guaranteed number of serviceable fibers.

9.3 TWISTED-PAIR (FOUR PAIR) DATA CABLE TESTS

A. The most common tools for UTP copper cable testing are the Level 3, or better, Electronic Field Testers for Category 6A cables. Level 3 testing shall be performed using a Fluke DSP 4000 or Owner-approved equal tester.

B. Method:

1). Prior to testing, ensure each cable has had any static charge dissipated. Use an 8 conductor grounding cable and plug the RJ-45 end to each jack location to drain any charge.

2). Follow TIA/EIA 568.B procedures for Category 6A station cable test. Test for NEXT, FEXT, ACR, Attenuation, continuity, power sum cross talk, Return Loss, length, etc. as specified by TIA/EIA. Any cable that cannot meet minimum performance criteria in TIA/EIA procedures must be replaced and retested to verify compliance, at no expense to the owner. Detailed, printed, and electronic copy of test results for each cable tested must be provided to the Owner upon completion of testing.

9.4 LARGE-COUNT TWISTED-PAIR PHONE CABLE TEST

A. Provide separate post-installation test schemes for the following activities: 1).

Interbuilding (Campus Subsystem) twisted-pair segment test plan. 2). Riser

(Vertical Subsystem) twisted-pair segment test plan.

3). Horizontal (Horizontal Subsystem) twisted-pair segment test plan.

B. The most common tools for UTP copper cable testing are Digital VOM, continuity tester, TIA/EIA Level 3 tester and time domain reflectometer (TDR).

1). Method: After terminating and/or splicing the cables. Test all cable pairs for continuity, ground fault, shorts and crossed pairs. Test the sheath on all metallic sheath cable for continuity, and proper grounding.

2). Use a Digital VOM to test:

a). Cable sheath for continuity, and proper grounding. b).
Each cable pair for the absence of ground.

3). Use a continuity tester or TIA/EIA Level 3 tester to test each cable pair for continuity, shorts and crossed pairs.

4). Use the TDR to pinpoint the location of shorts, grounds, or breaks in continuity.

C. For 25 pair or smaller replace entire cable if bad pair or conductor is found. For larger pair, count cables, replace if more than 1 percent of pairs are bad. All bad pairs must be clearly identified on the submitted test report. Correct any crossed, shorted, or reversed pairs.

PART 4 COMMUNICATIONS CABLE AND ACCESSORIES

GENERAL

10.1 SUMMARY

A. Sections Included:

1). This Section covers work necessary to furnish and install communications cable and accessories including the following:

a). Unshielded twisted-pair (UTP) cable. b). Fiber optic cable.
c). Composite horizontal cable. d). Multi-conductor connectors. e). Fiber optic connector.
f). Ground wire.
g). Shielded twisted pair (STP) cable.

10.2 SUBMITTALS

A. Provide applicable portions of the following information in addition to the contract requirements within 10 days after award of Subcontract:

1). Cable termination and test plans. 2).

Cable and accessory data sheets.

PRODUCTS

11.1 MULITI-PAIR UNSHIELDED TWISTED-PAIR BACKBONE CABLE

A. Acceptable Manufacturers:

1). Berk-Tek

B. Refer to Cable Schedule and Project Drawings for project-specific cable types.

C. Multi-pair UTP Category 3 Backbone Cable: Shall be 100 ohm UTP cable for use in the backbone distribution subsystem. Shall meet or exceed the requirements of EIA/TIA 568 B, 100 ohm UTP Multi-pair Backbone Cable.

1). Pair Size: As shown on the Cable Schedules.

2). Conductors: WECO color-coded 24 AWG solid copper thermoplastic-insulated conductors twisted together in pairs.

3). Conductor Core: Pairs twisted together into binder groups of 25 pairs and assembled to form a single compact core surrounded by a plastic core wrap.

4). Sheath:

a). Gel-Filled Cables: Cores are overlaid with corrugated aluminum and steel shields covered by a polyethylene outer jacket and filled with water blocking gel.
b). Air-Filled Riser Cables: Cores are overlaid with or without corrugated aluminum shield covered by a PVC outer jacket. Cables without shields extending outside of the Communications Room require a parallel external 6 AWG ground conductor.
c). Air-Filled Plenum Cables: Cores with or without aluminum shield overlay covered by a thermoplastic outer jacket suitable for use in plenums. Cables without shields extending outside of the Communications Room require a parallel external 6 AWG ground conductor.

5). Types:

a). Category 3 Riser Rated Cable:
1. Pair Size: As shown on the Cable Schedules.
2. Jacket: Semi-rigid PVC or FEP.
3. Unshielded.
4. Air filled.
5. Conductor: Solid 24 AWG.
6. ANSI/TIA/EIA 568A Cat. 3, UL Listing: CMR.
7. Use: In Comm. Rooms or to distribution points. b).

Category 3 Plenum Rated Cable:
1. Pair Size: As shown on the Cable Schedules.
2. Jacket: Semi-rigid PVC.
3. Unshielded.
4. Air filled.
5. Conductor: Solid 24 AWG.
6. ANSI/TIA/EIA 568A Cat. 3, UL Listed: CMP.
7. Use: In Comm. Rooms or to distribution points. c).

ARMM Riser Cable:
1. Pair Size: As shown on the Cable Schedules.
2. Jacket: PVC.
3. Shielded.
4. Air filled.
5. Conductor: Solid 24 AWG.

6. ANSI/TIA/EIA 568A Cat. 3, UL Listed: CMR, NEC Article 800 Riser.
7. Use: Between Comm. Rooms and/or Comm. Cabinets

D. OSP, UTP Cable for Aerial, Direct Burial or Underground Conduit Installation: 1).

Pair Size: As shown on the Cable Schedules.

2). Jacket: Semi-rigid PVC. 3).

Shielded.

4). Jell Filled.

5). Conductors: Solid 24 AWG.

6). NEC/UL Listing: None; not approved for general use within building except when installed in metallic conduit.

11.2 UNSHIELDED TWISTED-PAIR HORIZONTAL CABLE

A. Acceptable Manufacturers:

1). Berk-Tek LANmark-10G2 Plenum CAT6A UTP, part #10137694

B. UTP Horizontal Cable: 100 ohm UTP cable for use in the horizontal distribution subsystem shall meet or exceed the requirements of EIA/TIA 568 A 10.2, 100 ohm UTP Cable.

1). EIA/TIA Category 6A as shown on the Horizontal Cable Schedules. 2).

Conductors: WECO color-coded 23 AWG solid copper thermoplastic-insulated conductors twisted together in pairs.

3). Rated 140 degrees F and 300 volts. UL Listed as CMP-LP (0.5A) 4).

Construction: Bare copper wire insulated with FEP. Two insulated conductors twisted together to form a pair and four such pairs cabled to form the basic unit made round with 3 monofilaments and with a striated flame-retardant PVC jacket.

5). Four-pair cable sheaths must be clearly factory labeled at intervals not greater than 2 feet to easily identify Category, Manufacturer, NEC/UL Listing, AWG and distance marking.

C. Type UTP, Category 6A Four-Pair UTP Non-plenum Cable: NEC/UL Listing: CMR.

D. Type UTP, Category 6A Four-Pair UTP Plenum Cable: NEC/UL Listing: CMP.

11.3 SINGLE-MODE FIBER OPTIC CABLE

A. Acceptable Manufacturers:

1). Corning.

2). Berk-Tek.

B. Single-Mode Optical Fibers: 8.3/125 micron Class IVa dispersion-unshifted optical fibers for use in the backbone distribution subsystem shall meet or exceed the requirements of EIA, GIGABIT ETHERNET, ANSI X3T9.5 PMD and TIA/EIA 568 B. including the following specifications:

1). Chromatic Dispersion:

a). Zero-Dispersion Wavelength: Between 1,300 nm and 1,324 nm. b). Maximum value of the dispersion slope at zero dispersion wavelength shall be no greater than 0.093 ps per km nm2.
c). Mode Field Diameter: Nominal diameter shall be 8.7 microns to 10 microns, with a tolerance of plus or minus 0.5 micron at 1,300 nm.

2). Maximum Attenuation:

a). 0.5 dB per km. at 1,310 nm. b).
0.4 dB per km. at 1,550 nm.
c). Cutoff Wavelength of Cabled Fiber: Less than 1,270 nm.

3). Marking: Cable sheaths must be clearly factory labeled at intervals not greater than 2 foot to easily identify Manufacturer, Type, NEC/UL Listing, and distance marking.

C. Type 8.3/250, Backbone Single-Mode Fiber Optic Cable for Aerial, Direct Burial or Underground Conduit Installation:

1). Individual Fibers: 8.3/125/250 microns.

2). Assembly: Nonmetallic jell-filled, loose-tube, fiber core with dielectric strength member enclosed by a nonmetallic or metallic cross ply sheath (as required for application); requires buffer tubing.

3). NEC/UL Listing:

a). Outdoor rated: None; not approved for general use within building except when installed in metallic conduit.
b). Indoor/Outdoor rated: OFNR.

D. Type 8.3/900, Backbone Single-Mode Fiber Optic Cable for Indoor Installation: 1).

Individual Fibers: 8.3/125/250/900 microns.

2). Assembly: A nonmetallic sheath shall surround a core of individually tight-buffered fibers.

3). NEC/UL Listing: OFNR or OFNP as required for application.

11.4 MULTIMODE FIBER OPTIC CABLE

A. Acceptable Manufacturers:

1). Corning.

2). Berk-Tek.

B. Multi-mode Optical Fibers: 50 micron, Lazer-optimized optical fibers for use in the backbone and horizontal distribution subsystems shall meet or exceed the requirements of EIA GIGABIT ETHERNET, ANSI X3T9.5 PMD and EIA/TIA 568 B, including the following specifications:

1). Maximum Mean Fiber Loss: a).

3.0 dB per km at 850 nm.
b). 1.0 dB per km at 1,300 nm.

2). Minimum Band Width:

a). 700 Mhz per km at 850 nm. b).
500 Mhz per km at 1,300 nm.

3). Marking: Cable sheaths must be clearly factory labeled at intervals not greater than 2 foot to easily identify Manufacturer, Type, NEC/UL Listing, and distance marking.

C. Type 50 micron, Backbone Multi-mode Fiber Optic Cable for Aerial, Direct Burial, or Underground Conduit Installation:

1). Individual Fibers: 50/125/250 micron.

2). Assembly: Nonmetallic jell-filled loose-tube fiber core with dielectric strength member enclosed by a nonmetallic or metallic sheath (as required for application); requires buffer tubing.

3). NEC/UL Listing:

a). Outdoor Rated: None; not approved for general use within building except when installed in metallic conduit.
b). Indoor/Outdoor Rated: OFNR.

D. Type 50 micron, Multi-mode Fiber Optic Cable for Indoor Installation: 1).

Individual Fibers: 50/125/250/900 micron.

2). Assembly: A nonmetallic sheath surrounds Core of individually tight-buffered fibers.

3). NEC/UL Listing: OFNR or OFNP as required for application.

11.5 Composite FIBER OPTIC Cable for Aerial, Direct Burial or Underground Conduit Installation

A. Acceptable Manufacturers:

1). Corning.

2). Berk-Tek.

B. Specifications: The same as 2.3.B and 2.4.B above.

C. Type: Composite cable consisting of a minimum of 12 single mode 8.3/125/250 micron fibers and 12 multi mode 50/125/250 micron fibers.

D. Total count of each fiber type as shown on the Cable Schedules and Project Drawings.

E. Assembly: The same as 2.3.C and 2.4.C above.

11.6 COMPOSITE FIBER OPTIC CABLE for Indoor Installation

A. Acceptable Manufacturers:

1). Corning.

2). Berk-Tek.

B. Specifications: The same as 2.3.B and 2.4.B above.

C. Type: Composite cable consisting of a minimum of 12 single mode 8.3/125/900 micron fibers and 12 multi mode 50/125/900 micron fibers.

D. Total count of each fiber type as shown on the Cable Schedules and Project Drawings.

E. Assembly: The same as 2.3.D and 2.4.D above.

11.7 FIBER OPTIC CONNECTORS

A. Acceptable Manufacturers:

1). Corning.

B. Mechanical connectors are not permitted.

C. 8.3/125 Micron Single-Mode SC Connectors: Ceramic ferrules with rear body suitable for Loose Tube 3 mm maximum cable diameter; adhere to optical glass using epoxy or hot-melt adhesive.

D. 50/125 Micron Multi-mode SC Connectors: Ceramic ferrule with rear body suitable for Loose Tube 3 mm maximum cable diameter; adherence to optical glass using epoxy or hot-melt adhesive.

E. 50/125 Micron Multi-mode SC Connectors: Ceramic ferrule with rear body suitable for Tightbuffered 3 mm maximum cable diameter; adherence to optical glass using epoxy or hot-melt adhesive.

11.8 FIBER OPTIC SPLICE

A. Acceptable Manufacturers:

1). Corning.

B. Only aerial, direct burial, or underground conduit installed fiber optic cable may be spliced when:

1). Cable run distance exceeds maximum cable reel distance. 2). A

transition from outdoor to indoor cable is required by code.

3). Cable is damaged and a splice is required to restore service. 4).

When LC factory pigtails need to be connected to fiber cable.

5). When angle polished connectors need to be connected to fiber cable.

C. Method:

1). Only fusion splices are allowed. No mechanical splices.

2). Splice case must be rated for the environment and location used.

11.9 MULTI-PAIR UTP CABLE SPLICE

A. Acceptable Manufacturers:

1). 3M-710

B. Only aerial, direct burial, or underground conduit installed outside plant UTP cable may be spliced when:

1). Cable run distance exceeds maximum cable reel distance. 2). A

transition from Outdoor to Indoor cable is required by code. 3).

Cable is damaged and a splice is required to restore service.

C. Method:

1). Only multi-pair (25 pair) splice connectors may be used. No single pair or single conductor splices allowed.

2). Splice case must be rated for the environment and location used. No direct buried splices. All splice cases shall be placed in a manhole, handhole or building.

EXECUTION

12.1 FIBER OPTIC CABLE TERMINATION PLAN

A. Submit a detailed plan for fiber optic cable termination, describing the proposed methods for installing the fiber optic cables. The termination plan shall include how each system component will be installed at FDUs, proposed connector products, methods for terminating fiber strands, buffer tubing for 50/125/250 micron strands, and provisions for protecting and storing service tails.

12.2 PREPARATION

A. Verify that installed raceways, cable tray, equipment cabinets, equipment racks, cross-connect systems, cabling, and local area network systems are suitable for use and ready to receive work.

B. Use mandrel to clean and thoroughly swab conduit systems before installing cable.

C. Record and report to the Contractor any of the following conditions:

1). Sharp edges, burrs, or fasteners in raceway or cable trays. 2).

Cable tray elbows, or tees with radii of less than 12 inches.

3). Conduit elbows with an inside radius of less than 10 times the conduit diameter.

4). Wireways with sharp 90 degree elbows or tees.

5). Raceways or cable trays at or exceeding rated fill capacity. 6).

Incomplete sections of raceway or cable tray.

7). Pull boxes shall not be used as change of direction without Architect/Engineer and Owner approval prior to pull boxes installation.

D. Install cable in raceway and cable tray after interior of building has been physically protected from the weather.

12.3 GENERAL COMMUNICATIONS CABLING METHODS

A. Provide pull tape (mule-tape) with preprinted foot markers is usually provided when conduit and innerduct are installed. Provide pull tape in each empty communications conduit over 10 feet in length.

B. Cables shall be placed in raceway or cable tray in the following manner:

1). Multi-conductor cables of 25 pair or more shall not be placed on top of or crossing over Category 6A UTP or fiber optic cables.

2). Fiber optic cables shall be placed in inner-duct unless they are of an interlocking armored construction.

3). All cables shall be placed parallel to the direction of the raceway or cable tray.

4). Cables shall be evenly distributed across the raceway or cable tray. Bunching or bundling of cable shall not be allowed.

5). Loops or excess cable length shall not be stored in the raceway or cable tray.

6). UTP shall be secured to the cable tray at all elevations, horizontal and vertical bends by means of Velcro ties for Category 6A cables.

7). Splicing of copper or fiber optic cables shall not be allowed in cable tray or indoor pullboxes, unless approved in writing by Owner or A/E.

C. Where the termination of cables installed under this Section is to be made by others, provide labeled lengths adequate to allow for neat, managed routing to the termination points.

D. Splicing of copper or fiber optic cables shall not be allowed except as described in paragraphs 2.8 and 2.9 above.

E. The number of twisted pairs or fiber strands specified for each cable shall be terminated and working.

F. Provide suitable cable slack in vaults, maintenance holes, boxes, and outlets to ensure that there is no kinking or binding of the sheath. See paragraph 3.5 for specific guidelines.

G. Cable Management Mountings, Hangers, and Attachments:

1). Install straps, mountings, hangers, J hooks, support wires, distribution rings, cable tie mountings, and other similar fittings adequate to support loads with ample safety factors and so as not to damage, bind, or deflect cable. No greater than 5 feet between supports.

2). Plumbers perforated straps shall not be used as a means of support at any time.

3). Communications cable supports shall not be fastened to fire sprinkler, steam, water, or other piping, ductwork, mechanical equipment, electrical equipment, electrical raceways, or wires used to support ceiling grid.

4). Do not allow communications cables to make contact with ceiling tiles, ceiling grid support wires, or lighting fixtures.

5). Cable placed in walls shall be in conduit, 3/4 inch minimum diameter.

H. Communications Cable Ties:

1). Install cable ties loosely so as not to bind the cable sheath or distort the placement of twisted-pair components.

2). Plenum Building Areas: Where cable passes through open return air space, use plenum-rated plastic or stainless steel.

3). Non-plenum Building Areas: Use conventional flame-retardant nylon ties with a consistent natural or white color.

4). Exterior and Underground Locations: Use black plastic or stainless steel cable ties.

5). For Category 6A cables only use Velcro ties.

I. Communications Jumpers and Patch Cords: Route and arrange cross connects neatly in the wire management trays and troughs provided as part of the cross-connect system utilizing required standard length jumpers.

J. Provide the following minimum separation distances between pathways for communications cables with copper conductors and power wiring:

1). Open or Nonmetal Communications Pathways:

a). 12 inches from electric motors, fluorescent light fixtures, and unshielded power lines carrying up to 5 kVA.
b). 24 inches from electrical equipment and unshielded power lines carrying more than 5 kVA.

2). Grounded Metal Conduit Communications Pathways:

a). 2-1/2 inches from electrical equipment and unshielded power lines carrying up to 2 kVA.
b). 6 inches from electrical equipment and unshielded power lines carrying from 2 kVA to 5 kVA.
c). 12 inches from electrical equipment and unshielded power lines carrying more than 5 kVA.
d). 3 inches from power lines enclosed in a grounded metal conduit (or equivalent shielding) carrying from 2 kVA to 5 kVA.
e). 6 inches from power lines enclosed in a grounded metal conduit (or equivalent shielding) carrying more than 5 kVA.

3). Outdoor, aerial cable installations: Maintain a minimum of 40 inches separation below the lowest hung power or neutral conductor on a utility pole.

12.4 UNSHIELDED TWISTED-PAIR CABLE INSTALLATION

A. Place UTP cable so as to maintain the minimum cable bend radius limits specified by the manufacturer or the following, whichever is larger:

1). Horizontal Four-Pair UTP Cables:

a). Termination Points: Eight times the cable diameter. b). Other Locations: Four times the cable diameter.

2). Multi-pair UTP Cables: Maintain a minimum bend radius of ten times the cable diameter.

B. To avoid stretching four-pair horizontal cable conductors during installation, do not exceed a 25 pound force pulling tension (tensile loading).

C. Use wiring block and/or connector manufacturer's recommended tools, with the proper sized anvils, for copper punch-down, wire-wrap, and crimp terminations. Stuffer caps are not permitted, as a means for conductor punchdown.

D. Directly terminate twisted-pair cable on wiring blocks in standard WECO color- code order.

E. Cable Jackets: To reduce untwisting of pairs, maintain the twisted-pair cable jacket as close as possible to the point of termination.

1). Multi-pair Cable: Strip back only as much cable jacket as is required to terminate on connecting hardware.

F. Horizontal Cable: Strip back no more than 1 inch of cable sheathing.

G. Pair Twist: Preserve wire pair twists as closely as possible to the point of mechanical termination to minimize signal impairment; limit untwisting as follows:

1). Multi-pair Cables: Not greater than 1 inch.

2). Category 3 Horizontal Cables: Not greater than 1 inch.

3). Category 6A Horizontal Cables: Not greater than 1/2 inch.

12.5 FIBER OPTIC CABLE INSTALLATION

A. Use care when handling fiber optic cable. Carefully monitor pulling tension and cable bend radius so as not to exceed the limits specified by the manufacturer.

B. Maintain minimum of 12-inch bending radius for all fibers when pulling cable.

C. Fiber optic cables shall not be spliced except at locations indicated in paragraph 2.8 above.

D. Provide the following lengths of unsheathed fiber tail to allow for connectors and polishing.

1). FDUs: 15 feet stored in FDU. 2).

T.O.s: 4 feet stored in T.O.

E. Provide the following lengths of unsheathed fiber slack for repair and emergency restoration:

1). At the FDU each cable shall have an additional 20 feet of fiber. Neatly wrap the excess and locate in ceiling space above cable tray.

2). Maintenance Holes and Hand Holes: At locations indicated on Project Drawings, place a 20-foot slack coil. Secure the coil so will be out of the way of future installations. There should be enough locations in an underground conduit system to provide 10 percent overall slack.

3). Aerial Installations: At locations indicated on Project Drawings, place a “Snow Shoe” slack loop of 120 feet. There should be enough locations in an aerial system to provide 10 percent overall slack.

F. Fiber Optic Cable Termination:

1). Tight Buffer Strands: Directly terminate tight-buffered 8.3/125/900 micron single-mode or 50/125/900 micron multi-mode fiber optic cable strands at both the FDU and T.O. ends with SC connectors.

2). Loose Tube Strands: Install buffer tubes on loose tube 8.3/125/250 micron single-mode or 50/125/250 micron multi-mode fiber optic cable strands before directly terminating with SC connectors. Fusion spliced pigtails are required on single mode cable when cable length exceeds 15,000 feet or as shown on Project Drawings.

3). Buffered Strands: Directly terminate buffered 50/125/900 micron multi- mode fibers included in two-strand cable at both the FDU and T.O. ends with SC.

12.6 FIELD QUALITY CONTROL

A. Inspect wire and cable prior to terminations for physical damage and proper connection.

B. Verify cable is terminated according to the cable schedules.

C. Shield grounding, cable support, and placement are properly completed.

D. Testing of cable per provisions in “Communications Cable Testing Methods” section of this document.

12.7 PROTECTION

A. Protect existing cabling from damage by work in progress.

PART 5 COMMUNICATIONS IDENTIFICATION

GENERAL

13.1 SUMMARY

A. Section Includes:

1). This section specifies the requirements necessary to furnish and install communications labeling and marking including:

a). Conduit.
b). Cable.
c). Equipment.
d). Pull Box Enclosures.
e). Racks and Cabinets. f).
Innerduct.
g). Communication Rooms.

13.2 SUBMITTALS

A. Provide the following within 30 days after Award of Contract:

1). Submittals of materials, printing, and engraving methods.

B. Provide the following prior to the start construction:

1). Submittals of temporary construction labeling methods and materials.

PRODUCTS

14.1 MAN HOLES AND HANDHOLES

A. Labeling Method:

1). Man holes and handholes are identified with a unique number that describes the use as communications and includes a unique identifier.

2). Identification shall use the AABBB format, where:

a). AA = TM for man hole or TH for handhole. b).
BBB = Unique identification number, e.g.
1. Examples: TH001 identifies communications man hole or handhole number 001.

B. Labeling Method For Existing Man Holes and Handholes:

1). Existing man holes and handholes that are identified with a “Confined Space” number are valid. The existing “Confined Space” number shall be represented on the project documents for reference.

C. Material:

1). Welding electrode on the metal cover. 2).

Font Size: 4 inches.

D. Location: At the center of the metal cover.

14.2 SITE DUCT CONDUITS

A. Labeling Method:

1). Site duct conduits shall be identified with a unique string, which references the raceway type, start point, end point, and raceway sequence number.

2). Identification shall use the XX-BBBBB-CCCCC-EE format, where: a).

XX = Raceway type according to the following:

XX	Type
TC	Communication conduit

a). BBBBB = Start point identifier. c).
CCCCC = End point identifier.
d). EE = Raceway sequence number.
1. Examples: TC-TH001-TH002-01 identifies the first telecommunications conduit from man hole or handhole TH001 to man hole or handhole TH002.

B. Material:

1). Use a rigid, laminated, engraved plastic sign material. a).

Font: Arial.
b). Font Size: 3/8 inch. c).
Font Color: Black.
d). Background Color: Yellow. e).
Dimensions:
1. Length: 6 inches.

2. Width: 1 inch.

C. Mounting Methods:

1). Attach to the wall of the man hole/handhole, 1/2 inch above the duct entrance location, with 2 No. 10 screws, concrete anchors, and flat washers.

2). Adhesives and power-driven nails shall not be allowed.

3). When conduits enter man hole/handhole via bottom of box, label shall be attached directly to conduit with 2 bands of stainless steel.

14.3 Conduits

A. Labeling Method:

1). 2 inch and larger conduits shall be identified with a unique string, which references the raceway type, start point, end point and raceway sequence number.

2). Identification shall use the XX-BBBBB-CCCCC-EE format, where: a). XX = Raceway or cable tray type according to the following:

XX	Type
TC	Communication conduit

b). BBBBB =Start point c).
CCCCC = End point
d). EE = Raceway sequence number.
1. Examples: TC-TH001-TB002-01 identifies the first telecommunications conduit from telecommunications hand hole/man hole 001 to telecommunications pull box 002.

B. Material:

1). Use a flexible plastic, self-adhesive preprinted tape type sign material. Brady Corp, Labelizer® Plus Industrial Labeling System or equivalent.

2). Font: Arial.

3). Font Size: 1/2 inch.

4). Font Color: Black.

5). Background Color: Yellow.

6). Dimensions:

a). Length: 9 inches.
b). Width: 1 inch.

C. Location:

1). Labels shall be placed at the following locations:

a). At the start and end of that portion of the conduit system as per Drawings.
b). At each pull box.
c). At a maximum of 40 feet between labels.
d). At each side of wall penetrations within 15 feet of wall.

D. Mounting Methods:

1). Firmly apply label, level, to clean, dry and oil free surface. Label shall be applied late enough in project installation as to avoid nicks, tears or peeling flaps. If any nicks, tears or peeling flaps are noted prior to Owner accepting occupancy, label shall be replaced at no additional cost to Owner.

14.4 WIREWAYS (Gutter, Wiremold)

A. Use a rigid, laminated, engraved plastic sign material under 14.4.

B. Labeling Method:

1). Wire ways shall be identified with a unique string, which references the raceway type and raceway sequence number.

2). Identification shall use the XX-EE format, where: a). XX = Wire way type according to the following:

XX	Type
TC	Communication conduit

b). EE = Raceway sequence number.

C. Material:

1). Use a flexible plastic, self-adhesive preprinted tape type sign material. Brady Corp, Labelizer® Plus Industrial Labeling System or equivalent.

2). Font: Arial.

3). Font Size: 3/4 inch.

4). Font Color: Black.

5). Background Color: Yellow. 6).

Dimensions:

a). Length: 6 inches. b).
Width: 2 inches.

D. Location:

1). Labels shall be placed at the following locations:

a). At the start and end of a portion of the wire way system as per Drawings.
b). At each wire way right angle intersection. c).
At a maximum of 40 feet between labels.
d). At each side of wall penetrations within 15 feet of wall.

E. Mounting Methods:

1). Firmly apply label, level, to clean, dry and oil free surface. Label shall be applied late enough in project installation as to avoid nicks, tears or peeling flaps. If any nicks, tears or peeling flaps are noted prior to Owner accepting occupancy, label shall be replaced at no additional cost to Owner.

14.5 PULL BOXES AND GUTTERS

A. Use a rigid, laminated, engraved plastic sign material under Section 5.

B. Labeling Method:

1). Pull boxes shall be identified with a unique string, which references the type and pull box sequence number.

2). Identification shall use the XX-EEE format, where: a).
XX = Wire way type according to the following:

XX	Type
TC	Communication conduit

b). EEE = Pull box sequence number.

C. Material:

1). Use a flexible plastic, self-adhesive preprinted tape type sign material. Brady Corp, Labelizer® Plus Industrial Labeling System or equivalent.

2). Font: Arial.

3). Font Size: 3/8 inch.

4). Font Color: Black.

5). Background Color: Yellow. 6).

Dimensions:

a). Length: As needed.
b). Width: 2 inch.

D. Location:

1). Labels shall be placed at the following locations: a).

At the center of pull box cover.

E. Mounting Methods:

1). Firmly apply label, level, to clean, dry and oil free surface. Label shall be applied late enough in project installation as to avoid nicks, tears or peeling flaps. If any nicks, tears or peeling flaps are noted prior to Owner accepting occupancy, label shall be replaced at no additional cost to Owner.

14.6 INNER-DUCT

A. Labeling Method:

1). Inner-duct identification shall follow color sequence as identified in the “Communications Interduct” section of this document.

14.7 COMMUNICATIONS ROOMS

A. Labeling Method:

1). Communication Rooms shall be identified with a unique string, which references the building, communication room orientation in the building, and level as indicated on drawings.

2). Identification shall use the BCC format: a).

B =Building.

b). CC = Communication room orientation in Building.

1. Examples: B01-01 identifies the Building Communication Room in Building 01.
2. B01-01=Building Communication Room (BCR)
3. B01-02=Satellite Communication Room (SCR)

3). The Communication Room door sign shall be engraved to read: a).

Line 1: COMMUNICATION ROOM

b). Line 2: BCC

B. Material: SEE THE ARCH SPEC THAT COVERS ROOM SIGNS

C. Location: SEE THE ARCH SPEC THAT COVERS ROOM SIGNS 1).

Signs shall be placed at the following locations:

a). At the center of room door.

b). Elevation to center of sign 5 feet above finished floor.

D. Mounting Methods: SEE THE ARCH SPEC THAT COVERS ROOM SIGNS

14.8 DATA CENTER RACK AND CABINETS

A. Labeling Method:

1). Communication racks and cabinets shall be identified with a unique number for that room, which references the row number in the room and the rack/cabinet identifier as indicated on drawings.

2). Identification shall use the RR format.

a). RR=Communication Room Rack/Cabinet identifier, such as 01 or 23.

3). The Telecommunication room rows shall have a chain hung engraved sign to read:

a). Line 1: ROW NN.

4). The Telecommunication racks and cabinets shall have a self adhesive label that shall read:

a). Line 1: RACK RR or CABINET RR

B. Row Material:

1). Use a rigid, laminated, engraved plastic sign material. 2).

Font: Arial.

3). Font Size: 2 inch.

4). Font Color: Black.

5). Background Color: Yellow. 6).

Dimensions:

a). Width: 4 inch.
b). Length: 6 inches.

C. Rack or Cabinet Material:

1). Use a flexible plastic, self-adhesive preprinted tape type sign material. Brady Corp, Labelizer® Plus Industrial Labeling System or equivalent.

2). Font: Arial.

3). Font Size: 1/2 inch.

4). Font Color: Black.

5). Background Color: Yellow. 6).

Dimensions:

a). Width: 1 inch.
b). Length: 3 inches.

D. Location:

1). Signs shall be placed at the following locations:

a). For Racks: At the center of the top angle, facing the front and back of the rack.

b). For Cabinets: At the center top of the front panel of the cabinet. c). For Rows: Above the first rack as you enter the row from the main aisle way.

E. Mounting Methods for rigid laminate:

1). Chain hung from bottom of tray, top of sign to be 3 inches from bottom of tray.

2). Adhesives, sheet metal screws, and self -tapping screws shall not be allowed.

F. Mounting Methods for Rack/Cabinet:

1). Firmly apply label, level, to clean, dry and oil free surface. Label shall be applied late enough in room installation as to avoid nicks, tears or peeling flaps. If any nicks, tears or peeling flaps are noted prior to room Owner occupancy, label shall be replaced at no additional cost to Owner.

14.9 VOICE WALL FIELDS

A. Labeling Method:

1). Backboards shall be identified using a unique string, which references the telecommunications room it is located and the terminated pair count.

a). Examples: B01-01-400PR identifies Telecommunications Room B01-01, 400 pair.

2). Each backboard shall have minimum of eight 66 style-wiring blocks. Blocks shall be configured to 4 columns of 2 blocks each column. Each column shall be identified using a unique string, which references the row number in the Telecommunications Room, the rack number, and the patch panel to which it terminates.

a). Example: R01R03A identifies Row 01, Rack 03, Patch panel “A”. 3).

Each column of 2 stacked 66 style wiring blocks shall be labeled using printed color coded tags starting at upper left corner with “01” and continuing sequentially down left side, ending the bottom pair number on the left side with number “48”. Numbering shall continue on upper right corner of column with “49”, continuing sequentially down right side, ending the bottom pair number on the right side with number “96”. Hand written tags shall not be allowed.

B. Backboard Material:

1). Use a flexible plastic, self-adhesive preprinted tape type sign material.

Brady Corp, Labelizer® Plus Industrial Labeling System or equivalent.

2). Font: Arial.

3). Font Size: 1/2 inch.

4). Font Color: Black.

5). Background Color: Yellow. 6).

Dimensions:

a). Width: 1 inch. b).
Height: 6 inches.

C. Wiring block Material:

1). Use a printed paper designation strip. 2).

Font: Arial.

3). Font Size: To fit. 4).

Font Color: Black.

5). Background Color: As per manufacturer.

6). Dimensions: Per 110 style manufacturer’s requirements or as purchased from the manufacturer.

D. Location:

1). Labels shall be placed at the following locations:

a). For backboard identification attach the label to the upper right corner of the backboard
b). For wiring block identification snap fit the label to the side of each block.

14.10 DATA PATCH PANEL

A. Labeling Method:

1). Hand written labels shall not be allowed.

2). The port identification shall utilize the numerical sequence 1,2,3…

B. Material:

1). Use a flexible plastic, self adhesive, preprinted sign material. 2).

Font: Arial.

3). Font Size: 3/4 inch

4). Font Color: Black.

5). Background Color: White.

6). Dimensions:

a). Width: As required. b).
Height: 1-3/8 inch.

C. Location:

1). Labels shall be placed at the following locations:

a). For Patch Panel Identification: At the upper left corner of the patch panel facing the front.

D. Mounting Methods:

1). Thoroughly clean the front surface of the patch panel to remove dirt and grease.

2). Attach labels to the front of the patch panel with self-adhesive backing.

14.11 PHONE CABLE PATCH PANEL

A. Labeling Method:

1). Patch panels shall be identified with a single letter designation starting with "A".

2). Hand written labels shall not be allowed.

3). The patch panel port identification shall utilize the XXYYZZ format for each port.

4). XX= Carrier / 01 or 02

5). YY= Slot / 01 to 10

6). ZZ= Port / 01 to 48

B. Material:

1). Use a flexible plastic, self adhesive, preprinted sign material. 2).

Font: Arial.

3). Font Size: 1/4 inch

4). Font Color: White.

5). Background Color: Yellow. 6).

Dimensions:

a). Width: As required. b).
Height: 3/8 inch.

C. Location:

1). Labels shall be placed at the following locations:

a). For Patch Panel Port Identification: Centered above each port in the row starting at the upper left corner of the first patch panel in the rack facing the front. When these patch panels carry over to another row, pick up with the next number in the sequence.

D. Mounting Methods:

1). Thoroughly clean the front surface of the patch panel to remove dirt and grease.

2). Attach labels to the front of the patch panel with self-adhesive backing.

14.12 TELECOMMUNICATIONS OUTLET (T.O.)

A. Labeling Method for T.O:

1). T.O. shall be identified with a unique string, which references the BCR, SCRs or SCCs. The T.O. port shall be labeled with a four digit port number; such as “0001”, “0002”, etc.

2). Hand written labels shall not be allowed. 3).

As shown on drawings.

B. Material: T.O. Label:

1). Use a preprinted laminated plastic label. 2).

Font: Arial.

3). Font Size: [1/4 inch].

4). Font Color: Black.

5). Background Color: White.

6). Dimensions:

a). Width: As required. b).
Height: [3/8 inch].

C. Material: T.O. Port Label

1). Use a preprinted laminated plastic label. 2).

Font: Arial.

3). Font Size: 1/4 inch.

4). Font Color: Black.

5). Background Color: White.

6). Dimensions:

a). Width: As required. b).
Height: [3/8 inch].

D. Location:

1). Labels shall be placed at the following locations:

a). For T.O. Identification: Install in the space provided by the manufacturer for that purpose.
b). For T.O. Port Identification: Centered above each port.

E. Mounting Methods:

1). Thoroughly clean the front surface of the patch panel to remove dirt and grease.

2). Attach labels to the front of the patch panel with self-adhesive backing.

14.13 FIBER OPTIC DISTRIBUTION UNITS

A. Labeling Method:

1). Each Fiber Optic Distribution Unit (FDU) shall be identified with a unique string. Within the Communication Room, and rack in which it is located the FDU shall receive a C. There shall also be a termination schedule label that identifies each cable terminated in the FDU with the far end Communication Room, row, rack, sequence number, cable run number, and end ports occupied.

2). Hand written labels shall not be allowed.

3). The FDU identification label shall be as shown on drawings.

B. Material for FDU:

1). Use a flexible plastic, self-adhesive preprinted tape type sign material. Brady Corp, Labelizer® Plus Industrial Labeling System or equivalent.

2). Font: Arial.

3). Font Size: 3/4 inch.

4). Font Color: Black.

5). Background Color: Yellow for single mode and orange for multi mode fiber.

6). Dimensions:

a). Width: As required.

b). Height: 1 inch.

C. Material for termination and port label:

1). Use a flexible plastic, self adhesive, preprinted sign material. 2).

Font: Arial.

3). Font Size: 1/4 inch

4). Font Color: Black.

5). Background Color: Yellow for single mode Orange for 62.5 micron multi mode fiber and Green for 50 micron multi mode fiber.

6). Dimensions:

a). Width: As required. b).
Height: 3/8 inch

D. Location

1). Labels shall be placed at the following locations:

a). The FDU identification label shall have just the sequence number in the Upper Left Corner of the front and rear of the chassis cover.
b). The termination identification label for each cable shall be directly in line with the first port panel for that cable and applied to the front cover.
c). The port termination identification label for fiber shall be directly beneath the port termination location on the fiber panel.
d). Apply a label to each cable prior to entering the FDU. Label should be easily viewable.

E. Mounting Methods:

1). Thoroughly clean the front surface of the FDU to remove dirt and grease.

2). Attach the FDU labels to the FDU with a two-part epoxy adhesive.

3). Attach cable labels to the front of the FDU with self-adhesive backing. 4).

Attach port labels to the front of the fiber panel with self-adhesive backing.

14.14 BACKBONE CABLE

A. Labeling Method:

1). Backbone cable shall be identified with a unique string, which references the near end and far end Communication Rooms, cable type, pair or fiber count and fiber type.

2). Identification shall use the BBBBBB-RRRRRR-TTCCCGG where: a).

BBBBBB = Near End Communication Room number.
b). RRRRRR = Far End Communication Room number. c).
TT = Backbone cable type.
d). CCC = Fiber count or copper pair count. e).
GG = Fiber type.

TT	BACKBONE TYPE
FF	Fiber
FC	Composite Fiber
CC	Multi-strand copper
CX	Coax
GG	FIBER TYPE
SM	Single Mode
MM5	50 micron Fiber
MM6	62.5 micron Fiber

f). Example: B01-01-B24NW1-FC-72SM/36MM indicates a cable, which starts in Communications Room B01-01 and ends in Communications Room B24NW1, composite fiber with 72 strands of single mode and 36 strands of multi-mode.
g). Example: B01-01-B01-02-CC400 indicates a cable, which starts in Communications Room B01NE1 and ends in Communications Room B01-02, multi- strand copper cable with 400 pairs.

B. Material:

1). Use a flexible plastic, preprinted laminated sign material. 2).

Font: Arial.

3). Font Size: 1/8 inch.

4). Font Color: Black.

5). Background Color: Yellow.

6). Dimensions:

a). Length: 4 inches. b).
Width: 1 1/4 inches.

C. Location:

1). Labels shall be placed at the following locations:

a). At the point of entering the wiring block at the near and far end Communication Rooms.
b). At each pull box, man hole and handhole.

D. Mounting Methods:

1). Attach to the cable with 2 nylon cable ties through 1/4-inch holes at each corner of the label.

2). Adhesives shall not be allowed.

14.15 HORIZONTAL CABLE

A. Labeling Method:

1). Horizontal cable shall be identified with a unique string, which references the Communication Rooms, T.O. number and port.

2). As shown on drawings

B. Material:

1). Use a flexible plastic, preprinted adhesive wrap-around wire label manufactured for the purpose of labeling UTP cable. Paper label shall not be acceptable.

2). Font: Arial.

3). Font Size: [1/8 inch]

4). Font Color: Black.

5). Background Color: White.

6). Dimensions: As required.

C. Location:

1). Labels shall be placed at the following locations:

a). Each cable shall be labeled within 6 inches of connection to T.O. b). Each cable shall be labeled within 6 inches of connection to wiring blocks, FDU's and patch panels.
c). Contractor shall install labels immediately after placing and terminating cables.
d). Handwritten labels are not acceptable.

D. Mounting Methods:

1). Wrap the cable with the label such that the print is clear and visible. 2). If

at any time, prior to close out, the cable tag becomes illegible or removed for whatever reason, Contractor shall immediately replace it with a duplicate pre-printed cable tag at the Contractors expense.

3). Cable tags shall be easily accessible, both physically and visually upon completion of job.

EXECUTION

15.1 ***NOT USED***

PART 6 COMMUNICATIONS ROOM DISTRIBUTION SUBSYSTEM

GENERAL

16.1 SUMMARY

A. Section Includes

1). Backboards.

2). Cabinets.

3). Equipment racks.

4). Equipment rack shelves.

5). Protector panels and protectors. 6).

Wiring block systems.

7). Modular patch panels.

8). Fiber optic distribution units (FDUs). 9).

Cable management accessories.

10). Communication rooms tray and raceways. 11).

Ground busbar hardware.

12). Equipment rack and cabinet ground bars.

16.2 SYSTEM DESCRIPTION

A. The Communications Room distribution subsystem refers to the passive equipment used to manage and terminate backbone and horizontal cabling for the purpose of distributing communications services. This subsystem includes installations in the Campus Communications Room (CCR), Building Communications Rooms (BCRs), Satellite Communications Rooms (SCRs), and Satellite Communications Cabinets (SCCs).

16.3 SUBMITTALS

A. Closeout Submittals:

1). Project Record Documents

a). Record location of termination backboards, racks, and cabinets showing cable tray.
b). MCR, BCR, SCR, and SCC layouts showing completions and as- built corrections.
c). Front of bay elevation Drawings showing completions and as-built corrections.
d). Provide records of frame and equipment rack addressing or numbering by frame and shelf.
e). Provide wiring block and FDU addressing or numbering lists as indicated on schedules.

PRODUCTS

17.1 GENERAL

A. Labeling Materials: Labeling materials, including label holders and label inserts, shall be provided under the "Communications Identification" section of this document.

17.2 COMMUNICATIONS TERMINATION BACKBOARDS

A. Termination backboards shall be fire-retardant plywood furnished in sheets measuring 4-feet (1200-mm) wide, 8-feet (2400-mm) long, and 3/4-inch (18.75-mm) thick. A minimum of two sheets shall be installed in each room where called for on the Project drawings.

17.3 FLOOR SUPPORTED EQUIPMENT RACKS

A. Acceptable Manufacturers:

1). CPI

B. Description:

1). Material shall be high strength, lightweight 6061-T6 aluminum.

2). Finish shall typically be clear or as shown on the Project drawings. 3).

The upright's depth shall be 3 inches wide.

4). The uprights shall be provided with EIA drilling for forty-five (7'), forty- eight (7'6"), or fifty-one (8'), 1-3/4-inch panels on the front and rear.
Typically the 7'6" rack will be used but refer to the Project drawings for height application.

5). Mounting holes EIA drilling for the 1-3/4-inch panels shall be tapped for No. 12-24 screw thread on a 5/8" 5/8" ½" hole pattern.

6). Rack widths shall be ether 19 or 23 inches. Typically the 19" rack shall be used but refer to Project drawings for width application.

7). Top rails shall be 11/2" by 11/2" by ¼" thick angle. 8).

Base rails shall be 31/2" by 6" by 3/8" thick angle.

17.4 EQUIPMENT RACK INSTALLATION ACCESSORIES

A. Acceptable Manufacturers:

1). CPI

B. Description:

1). Installation kit for concrete floor shall include: a).

Four 3/8-16 concrete anchors.
b). Four 3/8-16 x 3 1/2-inch studs. c).
Four 3/8-16 nuts.
d). Four 3/8 lock washers. e).
Four 3/8 fender washers.

2). Installation kit for raised floor shall include: a).

Four 5/8-11 concrete anchors.
b). Four 5/8-11 threaded rods. Refer to project drawing for length. c).
Twelve 5/8-11 nuts.
d). Eight 5/8-11 lock washers. e).
Four 5/8 fender washers.
f). Four steel brackets. Refer to Project drawings for length.

3). Line-up spacer kit for attaching two racks mounted side by side shall include:

a). Two spacer plates 1 inch by 3 inches by 1/8 inch. b).
Four 5/16-18 by 2-inch hex bolts
c). Four 5/16-18 hex nuts. d).
Eight 5/16 plain washers.

4). Top brace shall include:

a). Two swivel brackets. Refer to project drawing for description. b).
Two 1/2-inch - 13 x 1-1/4 inch hex bolts.
c). Two 1/2-inch - 13 hex nuts. d).
Two 1/2-inch plain washers.
e). 5/8-11 threaded rod. Length as required. f).
Two 5/8-11 nuts.
g). Two 5/8 flat washers.
h). Attachment options:

1. Rack to Concrete:
   a. One 5/8-11 concrete anchor.
   b. One 5/8-11 x 3-inch stud.
   c. One 5/8-11 lock washers.
   d. One 5/8 flat washers.
   e. Two 5/11-11 hex nuts.
2. Rack to Overhead Frame:
   a. One grid clamp kit.
   b. One 5/8-11 x 3-1/4 inch hex bolt.
   c. One 5/8-11 lock washer.
3. Seismic Zone 3 and 4 top rails shall be 11/2” by 2” by ¼” thick angle and provided with two 0.688-inch diameter holes on the top surface, 2.84 inches from the ends.

17.5 WALL-MOUNTED STATIONARY EQUIPMENT RACK

A. Acceptable Manufacturers:

1). CPI

B. Description:

1). Material shall be as high strength, lightweight 6061-T6 aluminum. 2).

Rack height shall be as shown on the Project drawings.

3). Finish shall typically be clear or as shown on the Project drawings. 4).

The upright’s depth shall be 3 inches wide.

5). The uprights shall be provided with EIA drilling for up to forty-one 1-3/4-inch panels on the front. Refer to Project drawings for application.

6). Mounting Holes: EIA drilling for the 1-3/4-inch panels shall be tapped for No. 12-24 screw thread on a 5/8” 5/8” ½” hole pattern.

7). Panel widths shall be either 19 or 23 inches . Refer to Project drawings for application.

8). Maximum weight capacity shall be 200 lb.

17.6 WALL-MOUNTED SWING GATE EQUIPMENT RACK

A. Acceptable Manufacturers:

1). CPI.

B. Description:

1). Material shall be high strength, lightweight 6061-T6 aluminum. 2).

Rack height shall be 35 inches.

3). Finish shall typically be clear or as shown on the Project drawings. 4).

The upright's depth shall be 3 inches wide.

5). The uprights shall be provided with EIA drilling for twenty 1-3/4-inch panels on the front and rear.

6). Mounting holes EIA drilling for the 1-3/4-inch panels shall be tapped for No. 12-24 screw thread on a 5/8" 5/8" ½" hole pattern.

7). Panel widths shall be either 19 or 23 inches. Refer to Project drawings for application.

8). The Rack shall have dual locking latches to keep gate securely closed. 9).

Maximum weight capacity shall be 150 lb.

10). Gate shall swing 180 degrees.

17.7 VERTICAL CABLE MANAGEMENT SECTIONS

A. Acceptable Manufacturers:

1). CPI

B. Description:

1). Vertical Cable Management Sections shall be cable manager section consisting of: formed aluminum troughs, lockable cable rings, pass through ports with protective edge guards, and assembly hardware.

2). Styles shall be dual or single sided. Refer to Project drawings for application.

3). Height shall typically be 7'6" high. Refer to the Project drawings for height application of 7' or 8' high units.

4). Width shall be by 3-5/8 inches or 6 inches wide. Refer to Project drawings for application.

5). Vertical cabling sections shall be capable of being attached to the web face of the equipment rack uprights, ether the right or left sides.

6). Dual sided sections shall have predrilled pass through ports with protective edge guard installed; 21/2" pass through ports for the 6" managers and 13/8" pass through ports for the 35/8" managers.

7). Single sided 6" wide sections shall have 4" by 5" finger duct installed the entire length on the flat face. Drill three evenly spaced, 21/2" holes through the finger duct and the manager, and then line the hole with protective edge guard.

17.8 STANDARD EQUIPMENT SHELVES

A. Acceptable Manufacturers:

1). CPI.

B. Standard equipment shelves shall be used for supporting active equipment up to 100 pounds for a single sided model or 250 pounds for dual sided model.

C. Styles shall be dual or single sided with vented bottoms. Refer to Project drawings for application.

D. Finish shall typically be clear or as shown on the Project drawings.

E. Material shall be:

1). Aluminum.
2). Thickness: .090 inch thick aluminum.
3). Height shall be 5.6 inches.
4). 19-inch or 23-inch models shall be for mounting to a 19-inch or 23-inch standard equipment rack or cabinet. Refer to Project drawings for application.

F. Equipment shelves shall be mounted to the rack or cabinet uprights at the 1-3/4-inch high panel mounting spaces. Mounting hardware shall be a minimum of 6 No. 12-24 screws for single sided and 12 No. 12-24 screws for dual-sided models.

17.9 HEAVY DUTY EQUIPMENT SHELVES

A. Acceptable Manufacturers:

1). CPI.

B. Heavy-duty equipment shelves shall be used for supporting active equipment up to 200 pounds.

C. Styles shall be dual sided.

D. Finish shall typically be clear or as shown on the Project drawings.

E. Material shall be:

1). Aluminum.

2). Thickness.090 inch thick aluminum. 3).

Height shall be 7.2 inches].

4). 19-inch or 23-inch models shall be for mounting to a 19-inch or 23-inch standard equipment rack or cabinet. Refer to Project drawings for application.

F. Heavy-duty equipment shelves shall be mounted to the rack or cabinet uprights at the 1-3/4-inch high panel mounting spaces. Mounting hardware shall be a minimum of 12 No. 12-24 screws.

17.10 WALL-MOUNTED PROTECTOR PANELS

A. Acceptable Manufacturers:

1). Porta Systems

B. Wall-mounted protector panel shall be provided with five-pin wide-gap gas tube surge protector modules to provide protection for communication equipment and circuits exposed to voltage surges and sneak currents. 110 style connectors shall be provided as part of the factory assembly in 25, 50 or 100 pair models at both the input and the output of the protector panel. The proctor panel shall be compatible with the individual protector units specified below.

C. The lockable-hinged metal enclosure shall be provided from the manufacturer as part of the protector panel assembly.

D. The protector panel shall be securely to a backboard.

17.11 ELECTRICAL PROTECTOR MODULES

A. Wide-gap gas tube surge protector modules with solid-state 20-ohm heat coils shall be used to provide electrical protection for communications equipment and circuits exposed to voltage surges and fault currents. These black, five-pin modules plug into the protector panels above.

17.12 WIRING BLOCK SYSTEMS

A. Acceptable Manufacturers:

1). Leviton

B. 110 Wiring Blocks: Fire-retardant molded plastic blocks suitable for wall or equipment rack mounting, arranged with horizontal index strips that secure and organize 25 cable pairs each.

1). 100 Pair with Legs: 110AW2-100.

2). 300 Pair with Legs: 110AW2-300.

3). 100 Pair without Legs: 110DW2-100.

4). 300 Pair without Legs: 110DW2-300.

C. 110 Connecting Blocks: One-piece fire-retardant molded plastic housings, suitable for fastening onto 110 wiring blocks, which contain solder-plated quick clips that cut conductor insulation as they are pushed on the block, capable of terminating 22 AWG through 26 AWG conductors.

1). Three Pair: 110C-3.

2). Four Pair: 110C-4. 3).

Five Pair: 110C-5.

D. 110 Wiring Block System Accessories

1). 188UT1-50 transparent label holder. 2).

88A2 retainers.

3). 110 label inserts.

17.13 CATEGORY 6A PATCH PANELS

A. Acceptable Manufacturers:

1). Leviton Network Solutions part # 6A586-U48

B. Patch panel shall be a factory assembled and tested to EIA/TIA-568B wiring standards.

C. Ports shall be RJ45 eight-position/eight-conductor compliant with a T568B configuration and 110 termination connector.

D. The quantity of ports shall be 48 per patch panel. Refer to Project drawings for applications.

E. Patch panels may be mounted on 19-inch equipment racks or cabinets fitted with 19-inch mounting rails.

F. Wall mounting of patch panels shall only be allowed it if the patch panel is mounted to brackets designed for and intended for the purpose of wall mounting patch panels. Refer to Project drawings for application.

G. Category 6A patch panels SHALL ONLY be used for Network Systems and shall have GREEN lettering on the face. Refer to Project drawings for application.

H. Voice patch Panels shall be 24 ports, factory assembled and tested to EIA/TIA- 569B wiring standards.

I. Wall mounting of voice patch panels shall only be allowed it if the patch panel is mounted to brackets designed for and intended for the purpose of wall mounting patch panels. Refer to Project drawings for application.

J. Factory single pair, with 50 pin Amp connector on back, with pins 4 and 5 active.

17.14 MODULAR PATCH PANEL SYSTEM

A. Acceptable Manufacturers:

1). Leviton Network Solutions part # 49255-H48

B. Modular patch panel system shall consist of an empty multi-port frame, either flat or angled, designed to support the following port and cable termination configurations. Refer to Project drawing for applications.

1). Single CAT6A compliant RJ45 jack module. 2).

One simplex SC fiber adapter.

3). One simplex ST fiber adapter. 4).

One duplex LC fiber adapter. 5).

One blank insert.

C. The modular patch panel shall be metal and intended for installation in 19-inch equipment racks or cabinets fitted with 19-inch mounting rails.

D. CAT6A module ports shall be RJ45 eight-position/eight-conductor compliant with a T568B configuration and tool-free termination.

E. The quantity of modules per panel shall be 12, 24, 48, or 72 providing for 12, 24, 48, or 72 ports respectively. Refer to Project drawings for applications.

F. Wall mounting of modular patch panels shall only be allowed it if the patch panel is mounted to brackets designed for and intended for the purpose of wall mounting patch panels.

17.15 RACK-MOUNTED FIBER OPTIC DISTRIBUTION UNITS

A. Acceptable Manufacturers:

1). Corning.

B. Fiber distribution unit shall consist of an enclosure that is compatible of accepting a verity of coupling plates to support the listed fiber optic cable termination configurations. Refer to project drawing for applications.

1). Two panels with six duplex SC connectors for up to 24 fibers. 2). Four

panels with six duplex SC connectors for up to 48 fibers. 3). Six panels

with six duplex SC connectors for up to 72 fibers.

4). Twelve panels with six duplex SC connectors for up to 144 fibers.

5). Blank panels as required, to fill unused locations.

C. Enclosure, coupling plates, and couplings shall be of the same manufacturer and design.

D. Finish color shall be black.

E. The quantity of fiber optic strands terminated in an enclosure shall be 24 to 144. Refer to Project drawings for applications.

F. The FDU enclosure shall be fitted with fiber optic strand management such that the specified bend radius of the fiber optic strands will be maintained in an orderly and controlled manner.

G. The front and rear of the enclosure shall be provided with hinged [latchable] doors to protect terminations and jumpers from direct physical damage.

H. The enclosure shall be metal and intended for installation in 19-inch equipment racks or cabinets fitted with 19-inch mounting rails.

17.16 WALL-MOUNTED FIBER OPTIC DISTRIBUTION UNITS

A. Acceptable Manufacturers:

1). Corning.

B. Wall-mounted fiber distribution unit shall consist of an enclosure that is compatible of accepting a verity of coupling plates to support the listed fiber optic cable termination configurations. Refer to project drawing for applications.

1). Two panels with six duplex SC connectors for up to 24 fibers 2). Four

panels with six duplex SC connectors for up to 48 fibers 3). Six panels

with six duplex SC connectors for up to 72 fibers.

4). Twelve panels with six duplex SC connectors for up to 144 fibers. 5).

Blank panels as required to fill unused locations.

C. Enclosure, coupling plates, and couplings shall be of the same manufacturer and design.

D. Finish color shall be black.

E. The quantity of fiber optic strands terminated in an enclosure shall be 12 to 144. Refer to Project drawings for applications.

F. The FDU enclosure shall be fitted with fiber optic strand management such that the specified bend radius of the fiber optic strands will be maintained in an orderly and controlled manner.

G. The front of the enclosure shall be provided with hinged [latchable] doors to protect terminations and jumpers from direct physical damage.

H. The enclosure shall be metal and intended for wall-mount installation.

17.17 WALL FIELD CABLE MANAGEMENT ACCESSORIES

A. Acceptable Manufacturers:

1). CPI.

B. Wall-mounted wire management shall be manufactured and intended to be installed as backboard wire managers. Wire managers shall be compatible with backboard-mounted wire blocks, patch panels and FDUs.

C. Open and closed Cable Distribution Rings styles shall be J-hooks, C-rings, and D-rings made of plated steel or plastic. Refer to Project drawings for applications.

D. Wire Distribution Post shall be a plastic spool/post for mounting on backboards with screw to hold jumper wires in place.

17.18 EQUIPMENT RACK/CABINET CABLE MANAGEMENT ACCESSORIES

A. Acceptable Manufacturers:

1). CPI

B. Rack/cabinet cable management-mounted wire management shall be manufactured and intended to be installed as wire and jumper managers. Wire managers shall provide an orderly means of routing wire and jumpers from any location on the frame to any location on the frame. Wire managers shall be compatible with mounted wire blocks, patch panels, and FDUs.

1). Small ring horizontal wire managers shall be one-position wire managers. The distribution rings shall be 1.75 inches high with rounded edges to protect the bending radius and sheath of cross connect wire, patch cords and fiber jumpers. 19-inch or 23-inch models shall be for mounting to a 19-inch or 23-inch standard equipment rack or cabinet.
Refer to Project drawings for application.

2). Large ring horizontal wire managers shall be two-position wire managers. The distribution rings shall be 3.5 inches high with rounded edges to protect the bending radius and sheath of cross connect wire, patch cords and fiber jumpers. 19-inch or 23-inch models shall be for mounting to a 19-inch or 23-inch standard equipment rack or cabinet. Refer to Project drawings for application.

17.19 GROUND BUSBAR HARDWARE

A. Acceptable Manufacturers:

1). CPI

B. Ground busbar shall be solid copper plate, predrilled for two-hole lug connections, with 1/4-inch mm) minimum thickness, for wall and backboard mounting using standoff insulators manufactured and intended for use with the provided busbar.

1). Sizes shall be 4-inch by 10-inch for SCRs or 4-inch by 20-inch for CCR and BCR. Refer to Project drawings for application.

C. Ground terminal blocks shall be screw lug-type terminal blocks. A ground terminal block shall be installed at any location where rack-type ground bars cannot be mounted and installed at any equipment mounting location (e.g., backboards and hinged cover enclosures).

17.20 EQUIPMENT RACK AND CABINET GROUND BARS

A. Acceptable Manufacturers:

1) CPI.

B. Ground bars shall be solid copper ground bars intended for use on equipment racks and cabinets with minimum dimensions of 1/4-inch thick by 3/4-inch wide. Lengths shall be compatible with 19-inch or 23-inch equipment racks and cabinets. Refer to project drawing for application.

EXECUTION

18.1 PREPARATION

A. Verify that surfaces are ready to receive work. Be sure that raceway and cable tray routes are free from debris or obstructions.

B. Verify that field measurements are as shown on the Project drawings.

18.2 INSTALLATION

A. Install communications backboards, equipment racks and cabinets, boxes, raceways, framing, cable tray, wiring blocks, FDUs, and other related materials and equipment plumb as shown on the Project drawings and in accordance with the manufacturer's specifications and procedures.

B. Finishes:

1). Communications plywood backboards shall be painted with durable, fire-retardant paint to match the wall color.

2). Racks, cabinets, panels, and accessories shall be as per this document or Project drawings if different.

C. Install surface-mounted raceways in accordance with the provisions of the “Communications Raceways and Boxes” section of this document.

D. Install cable tray in accordance with the provisions of “Communications Cable Trays” section and “Communications Supporting Methods” section of this document.

E. Provide split-corrugated inner-duct as specified in the “Communications Inner Duct” section of this document to protect fiber optic cables at transitions from cable tray systems to equipment cabinets and racks.

F. Place and terminate twisted-pair and fiber optic cable in accordance with the provisions of the “Communications Cross Connections” section of this document. The following additional requirements shall apply:

1). Cables shall be routed in overhead cable trays and inside cable management systems attached to the equipment cabinets and racks.

2). Use hook and loop (Velcro) ties or ducts to restrain cabling installed outside of cable management systems on racks or in cabinets.

G. Furnish and install wire and hardware required to properly ground and bond communications cable and equipment in accordance with the provisions of the “Communications Grounding” section of this document. The following additional requirements apply:

1). Install building and/or telecommunications ground busbar as shown on the Project drawings.

2). Connect a pigtail extension from the building grounding facility to the Communications Room ground bus hardware as shown on the Project drawings.

3). Install a solid copper ground bar at the rear of each equipment cabinet and connect to the telecommunications ground busbar using a minimum of 6-AWG green insulated copper wire.

4). Ground metallic raceways, cable trays, cable shields, equipment racks, equipment cabinets, and other metallic equipment located in a Communications Room to the telecommunications ground busbar using a minimum of 6-AWG green insulated copper wire.

H. Identify and label the system in accordance with the provisions of the “Communications Identification” section of this document.

PART 7 COMMUNICATION CROSS CONNECTIONS

GENERAL

19.1 SUMMARY

A. Section Includes:

1). This Section specifies the requirements necessary to furnish and install communications cross connections, including:

a). Twisted-pair jumper wire. b). Twisted-pair patch cords.

2). Fiber optic patch cords.

19.2 CLOSEOUT SUBMITTALS

A. Project Record Documents:

1). Provide as built wiring block, Fiber Distribution Unit (FDU), and/or T.O. addressing and numbering lists for cross-connect jumpers and patch cords installed.

2). Submit cross-connect schedule check-off list for each installed, terminated, and labeled patch cord showing completions and record corrections.

PRODUCTS

20.1 TWISTED-PAIR PATCH CORDS

A. Acceptable Manufacturers:

1). Leviton Network Solutions part # 6AS10-**G or HF6AU-**G (High Flex)

B. The Subcontractor shall supply a minimum of:

1). Three patch cords for each desk location, one each in the Communication Room for data and phone connections and one each at the desk location for data connection.

2). One for each wall phone location, one each in the Communication Room for phone connection.

3). Lengths shall be determined with Owner prior to ordering any patch cords but typical length shall be 7 feet.

4). Distinctive colors shall be used to denote different Systems and Category as indicated below.

C. Category 6A RJ45 modular plug to Category 6A RJ45 modular plug shall be a factory assembled eight-conductor patch cord with straight-through wiring for connectivity from patch panel to patch panel, patch panel to active equipment ports and active equipment to active equipment. These cords shall be green in color. Wiring configuration shall be 568-B.

20.2 FIBER OPTIC PATCH CORDS

A. Acceptable Manufacturers:

1). Corning.

2). Leviton.

B. SC connector to LC connector, 50/125 micron multi-mode fiber optic station cords shall be factory assembled and tested. There shall be 1 type of assembly supplied, two-fiber assembly.. Cords shall be supplied with SC style connector at one end and a LC connector at the other end of the assembly. Jumpers shall be aqua in color.

C. SC connector to SC connector, 50/125 micron multi-mode fiber optic station cords shall be factory assembled and tested. There shall be 2 types of assemblies supplied single- and two-fiber assemblies. Cords shall be supplied with SC style connector(s) at both ends of the assemblies. Jumpers shall be aqua in color.

D. SC connector to SC connector, 8.3/125 Micron Single-Mode Fiber Optic Patch Cords shall be factory assembled and tested. There shall be 2 types of assemblies supplied—single- and two-fiber assemblies. Cords shall be supplied with SC style connector(s) at both ends of the assemblies. Jumpers shall be yellow in color.

E. SC connector to LC connector, singlemode fiber optic station cords shall be factory assembled and tested. There shall be 1 type of assembly supplied, two- fiber assembly.. Cords shall be supplied with SC style connector at one end and a LC connector at the other end of the assembly.

20.3 CROSS-CONNECT JUMPER WIRE

A. Cross-connect jumper wire shall be installed for the purpose of providing connectivity from a 110 style block to a 110-style block. Two types of cross- connect jumper wire shall be supplied with 2 conductors. Cross-connect jumper shall only be installed to facilitate Category 3 data or voice applications. One roll will be colored as Yellow Blue/Blue Yellow (for analog) and one roll to be Red Blue/Blue Red (for digital)..

EXECUTION

21.1 PREPARATION

A. Verify that cable raceways, equipment cabinets, equipment racks, cross- connect systems, cabling, and local area network systems are ready to receive work.

21.2 INSTALLATION

A. Cross-connect systems as shown on the Drawings, Cable Schedule and according to manufacturer's installation instructions.

B. Route and arrange cross-connect jumper wire and patch cords neatly in the horizontal and vertical wire management provided as part of the communication room subsystem. Cut cross-connect jumper wire to proper length and remove surplus wire from frame. Excess patch cord length shall be placed in the vertical wire manager in a neat and orderly method.

C. Identify and label the system as specified in the “Communications Identification” section of this document.

21.3 FINAL TEST

A. Conduct segment tests on field-assembled twisted-pair and fiber optic patch cords in accordance with the provisions in the "Communications Cable Testing Methods” section as required.

PART 8 COMMUNICATIONS RACEWAYS AND BOXES

GENERAL

22.1 SUMMARY

A. Section Includes: This Section specifies the requirements to furnish and install communications conduit raceways and boxes including:

1). Communications conduit systems. 2).

Wireway systems.

3). Surface raceway systems.

4). Telecommunications outlet boxes. 5).

Pull box enclosures.

6). Hinged cover enclosures.

7). Cabinets.

8). Pull-tape and duct plugs.

9). Raceway identification banding.

B. Related Specifications: This section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for communications raceways and boxes:

1). 07270—Firestopping.

2). 16130 – Raceways and Boxes.

3). 16135—Underground Raceways, Handholes, and Vaults.

22.2 DEFINITIONS

A. Cabinet: Freestanding floor-mounted modular enclosures designed to house and protect rack-mounted electronic equipment.

B. Conduit: Conduit is a round raceway.

C. Hinged Cover Enclosure: Wall-mounted box with a hinged cover that is used to surround electrical devices against specified external conditions and personnel against accidental contact with the enclosed devices.

D. Pull Box Enclosure: Box with a cover that is installed in one or more runs of raceway to facilitate pulling conductors through the raceway system. There are no openings in the cover. The cover may be hinged or multi screw connected to the box.

E. Raceway: Enclosed channel designed expressly for holding wires or cables. May be of metal or insulating material, and the term includes conduit, tubing, wireways, under-floor raceways, and surface raceways; does not include cable tray.

F. Surface Raceway: Surface-mounted metal channel or plastic duct with snap-in removable covers for housing and protecting electrical wires and cables. Raceway and fittings are designed so sections can be electrically and mechanically coupled together without subjecting cables to abrasion.

G. Wireway: Sheet metal or nonmetallic troughs with hinged or removable covers for housing and protecting electrical wires and cables and in which conductors are laid in place after the wireway has been installed as a complete system.

22.3 CLOSEOUT SUBMITTALS

A. Project Record Documents:

1). Plan Drawings showing completions and as built corrections that indicate type, size, placement, routing, and/or length for raceway components, e.g., maintenance holes, handholes, conduit, wireway, boxes, enclosures, etc.

PRODUCTS

23.1 GENERAL

A. Refer to Specifications 16130 and 16135.

23.2 RIGID METAL CONDUIT AND FITTINGS

A. Minimum size: Telecomm EMT shall be no smaller than ¾ inch and carry no more than 6 Category 5E cables or no more than 4 Category 6 cables.

23.3 ELECRICAL METALLIC TUBING AND FITTINGS

A. Minimum size: Telecomm EMT shall be no smaller than ¾ inch and carry no more than 6 Category 5E cables or no more than 4 Category 6 cables.

23.4 RIGID NONMETALLIC CONDUIT AND FITTINGS

A. Minimum size: Telecomm EMT shall be no smaller than ¾ inch and carry no more than 6 Category 5E cables or no more than 4 Category 6 cables.

23.5 ELECTRICAL NONMETALLIC TUBING AND FITTINGS

A. Type ENT: NEMA TC 12; corrugated flexible PVC tubing.

B. Fittings: In line straight-through terminal fittings identified for such use; do not use bends or tees, e.g., LBs.

C. Bushings: Nonmetallic.

D. Minimum size: Telecomm EMT shall be no smaller than ¾ inch and carry no more than 6 Category 5E cables or no more than 4 Category 6 cables.

23.6 LIQUID-TIGHT FLEXIBLE METAL CONDUIT AND FITTINGS

A. Minimum size: Telecomm EMT shall be no smaller than ¾ inch and carry no more than 6 Category 5E cables or no more than 4 Category 6 cables.

23.7 LIQUID-TIGHT FLIXIBLE NONMETALLIC CONDUIT AND FITTINGS

A. Minimum size: Telecomm EMT shall be no smaller than ¾ inch and carry no more than 6 Category 5E cables or no more than 4 Category 6 cables.

23.8 CONDUIT ACCESSORIES

A. Duct Plugs:

1). Aboveground Conduit Openings: Tapered PVC plugs with tab for pull tape.

2). Underground or Underslab Conduit Openings: Removable screw tight compression type duct plugs with wing nut and corrosion resistant hardware.

3). Empty conduits.

4). See “Communications Inner Duct” section for innerduct plugs. 5).

Duct Water Seal to be used with all duct plugs.

23.9 COMMUNICATIONS WARNING DEVICES

A. Tape: Heavy gauge, detectable orange plastic tape of 6-inch minimum width designed to be buried. Utilize tape made of material resistant to corrosive soil that includes metallic conductor suitable for detection by cable trace equipment. Use tape with printed warning that a telephone or fiber optic cable is located below the tape.

B. Stakes: 66”, orange, fiberglass, marking stake with black on orange 3” x 12” reflective decal stating “FIBER OPTIC CABLE BURRIED BELOW WARNING BEFORE DIGGING CALL”.

C. Approved Manufacturer:

1). VIP DIVISION, #VFS-66 – VMSD-FOR.

D. These are to be placed every between Handholes/Maintenance Holes at mid- point or every 135 feet, which ever is less.

23.10 RACEWAY COATING

A. Bitumastic material or plastic tape. To be used in corrosive or wet areas.

23.11 WIREWAY SYSTEMS

A. Minimum size shall be 6” x 6” x length as required.

23.12 CABINETS

A. Equipment Rack Enclosures (Freestanding Cabinets): Size, construction, finish, and configuration as shown on the Drawings.

EXECUTION

24.1 INSTALLATION

A. Refer to Specifications 16130 and 16135.

B. Minimum Conduit Size:

1). 3/4 inch for exposed conduit, 3/4 inch for aboveground concealed conduit, and 4 inches for underground applications unless otherwise indicated on the Drawings.

C. Conduit Bends and Sweeps:

1). The angular sum of the bends between pull points and/or pull boxes shall not exceed 180 degrees.

2). Minimum Inside Bend Radius for Communications Conduit Bends, Sweeps, Boxes, and Fittings:

a). Underground or Underslab 4-inch Conduit: 48 inches, unless noted otherwise.
b). Other Conduit Runs as noted below or closest next size larger factory bend:
1. One-inch (nominal size) conduit, 11 inches.
2. Two-inch (nominal size) conduit, 21 inches.
3. Three-inch (nominal size) conduit, 31 inches.
4. Four-inch (nominal size) conduit, 40 inches.
5. Other sizes, 10 times the inside diameter of the conduit.

3). Bends required for underground PVC shall be wrapped rigid galvanized steel.

D. Penetrations:

1). Seal conduit entering structures at the first box or outlet to prevent the entrance of gases, liquids, or rodents into the structure.

a). Empty Conduits: Removable screw-tight duct plugs.
b). Innerduct Installed: Suitable duct water seal between conduit and innerduct. Manufactured removable screw tight duct plug seals in empty innerduct.
c). Cable Installed: Suitable duct water seal between conduit and cable, or between innerduct and cable.

E. Above-Ground Conduit Installation:

1). No section of conduit located within buildings shall exceed 200 feet in length and the angular sum of the bends between pull points and/or pull boxes shall not exceed 180 degrees.

2). No section of conduit located exterior to building shall exceed 400 feet in length and the angular sum of the bends between pull points and/or pull boxes shall not exceed 180 degrees.

3). Flexible Conduit:

a). For final connection to Telecommunications Outlets or equipment, where flexible connection is required to minimize vibration or where required to facilitate removal or adjustment of equipment, provide 60-inch maximum lengths of flexible conduit as indicated on the Drawings.

F. Pull Tape and Duct Plugs:

1). Following conduit installation, install pull tape with preprinted foot markers in each empty conduit containing a bend or over 10 feet in length, except sleeves, nipples, and runs with openings in cleanroom areas. Tie the pull tapes securely to duct plug or wall racking at each end.

2). Immediately after pull tape installation, install removable manufactured plugs in empty conduit and wireway openings. For underground conduit openings, use screw tight, removable, watertight, and dust-tight duct plugs.

3). Verify lengths at the time of installation and provide as built documentation.

G. Grounding:

1). Provide ground connections and bonding continuity between raceway sections, boxes, enclosures, cabinets, and fittings in accordance with the "Communications Grounding" section.

PART 9 COMMUNICATIONS SUPPORTING METHODS

GENERAL

25.1 SUMMARY

A. Section Includes: This Section specifies the requirements to furnish and install communications supporting methods for the following:

1). Raceway, cable tray, and equipment supports. 2).

Fastening hardware.

B. Related Specifications: This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for communications supporting methods:

1). Specification 01446 - Seismic Restraints Criteria.

PRODUCTS

26.1 Material

A. Support Channel (Super-strut or equivalent): Steel, electrolytically coated with zinc; or fiberglass.

B. Threaded rod in various lengths.

1). 3/8 inch in diameter by 16 threads per inch. 2).

1/2 inch in diameter by 13 threads per inch. 3). 5/8

inch in diameter by 11 threads per inch.

C. Mounting, Attachment, and Anchoring Hardware: Corrosion resistant, stainless steel or non metallic.

D. Anchors, Nuts, Washers, and Bolts: Plated.

EXECUTION

27.1 APPLICATION

A. Install the following types of threaded rod where required:

1). Strut-Type Support Channels:

a). 3/8 inch in diameter by 16 threads per inch. b).
1/2 inch in diameter by 13 threads per inch.

27.2 INSTALLATION

A. All systems shall be plumb and level unless otherwise denoted on Project Drawings.

B. Fasten hanger rods, threaded rods, conduit clamps, and pull boxes to building structure using precast insert system, expansion anchors, beam clamps.

C. Use toggle bolts or hollow wall fasteners in hollow masonry, plaster, or gypsum board partitions and walls; expansion anchors or preset inserts in solid masonry walls; self-drilling anchors or expansion anchors on concrete surfaces; sheet metal screws in sheet metal studs. Do not use double-sided adhesive tapes.

D. Independently support raceways and cable trays from structural members only. Do not fasten raceway supports to pipe hangers or rods, piping, ductwork, mechanical equipment, electrical equipment, or raceway.

E. Anchor bolt installation in concrete decking shall be inspected by the Structural QC prior to them being used to support raceways and cable trays.

F. Do not drill structural steel members without written permission from Owner, A/E and Contractor.

G. Fabricate raceway and cable tray supports, such as ceiling trapeze, from structural steel or steel channel, rigidly welded or bolted to present a neat appearance. Use hexagon head bolts with spring lock washers under nuts. Attach raceway and cable tray to support structure using appropriate strap or clamp hardware.

H. Support and fasten each conduit with galvanized straps, lay-in adjustable hangers, clevis hangers, or bolted split clamps made of steel or malleable iron with rust-resistant zinc- or cadmium-plated finish.

I. Support and fasten each J-Hook support to 3/8 inch threaded rod with integral angle bracket. Independently support to structural members only. J-Hook shall have UL listed load rating. Use lock washers under hexagon nuts above and below angle bracket attachment to threaded rod.

J. Arrange conduit supports to prevent distortion of alignment by cable pulling operations.

K. Do not fasten raceways with wire or perforated pipe straps. Remove wire used for temporary raceway support during construction, before wire and cable are pulled.

L. Securely attach backboards, enclosures, and cabinets to wall or floor with a minimum of 4 anchors:

   1). Install material and trim plumb and level.

   2). Anchor surface-mounted enclosures using structural supports at each corner.

   3). Provide accessory feet for freestanding equipment cabinets.

M. Bridge studs top and bottom with channels to support recessed mounted enclosures in stud walls.

N. Use fiberglass or special coated metallic supports in areas subject to corrosives.

O. Support cable tray and multiple raceways adjacent to each other by ceiling trapeze.

P. Support individual raceways by wall brackets, strap hangers, or ceiling trapeze; fastened by wood screws on wood, toggle bolts on hollow masonry units, expansion shields on concrete or brick, and machine screws or welded thread studs on steelwork.

Q. Support exposed conduit with Super-Strut channels, Unistrut P 1000 channels, or equal, anchored to building construction with 1/4 inch Phillips Red Head anchoring device. Threaded studs driven in by a powder charge and provided with lock washers and nuts may be used in lieu of expansion shields where approved per paragraph 3.2E above.

R. Do not use nails anywhere or wooden plugs inserted in concrete or masonry as a base for raceway or box fastenings. Do not weld raceways or pipe straps to steel structures. Do not use wire in lieu of straps or hangers.

S. Install supports in accordance with the provisions of Specification 01049, Seismic Restraints Criteria.

PART 10 COMMUNICATIONS INNER DUCT

GENERAL

28.1 SUMMARY

A. Section Includes: This Section covers work necessary to furnish and install communications inner-duct and fittings.

28.2 SYSTEM DESCRIPTION

A. The inner-duct pathway system refers to the intra-building and inter-building paths for fiber optic communications cabling connecting site entrance locations, Campus Communication Room (CCR), Building Communication Rooms (BCR), Satellite Communications Rooms (SCR), and Satellite Communications Cabinets (SCC). Inner-duct shall be routed in conduits, underground duct banks, cable trays, and wire ways.

B. Inner-duct shall be routed and sized as per Inner-duct Schedules.

28.3 CLOSEOUT SUBMITTALS

A. Project Record Documents:

1). Submit inner-duct schedule check-off list for each installed and labeled inner-duct showing completions and as built corrections.

PRODUCTS

29.1 INNER-DUCT

A. Acceptable Manufacturers:

1). Duriline.

2). Carlon.

B. Outdoor Inner-Duct in Conduit: Smooth-walled outside with ribbed inside, semi- rigid PVC or heavy-wall polyethylene tubing.

1). Color: As indicated by size. 2).

Sizes shall be:

a). 3/4 inch inside diameter, in red, black, and gray.

b). 1 inch inside diameter, in blue and orange. c). 1
1/4 inch inside diameter, in white or yellow.

C. Indoor Inner-Duct in conduit: Smooth-walled outside with ribbed inside, semi- rigid nonflammable PVC tubing which meets UL 94V O vertical flame test for general applications.

1). Color: As indicated by size.

2). Sizes shall be:

a). 3/4 inch inside diameter, in red, black, and gray. b).
1 inch inside diameter, in blue and orange.
c). 1 1/4 inch inside diameter, in white or yellow.

D. Indoor Inner-Duct in Cable Tray: Corrugated walled inner-duct for use in riser or plenum air-handling spaces as required.

1). Color: Orange for multi-mode and yellow for single-mode. 2).

Sizes, as required, shall be:

a). 3/4 inch inside diameter. b).
1 inch inside diameter.
c). 1 1/4 inch inside diameter.

E. Split Corrugated Inner-Duct: Corrugated wall, semi-rigid, nonflammable PVC tubing which meets UL 94V O vertical flame test for general applications, with longitudinally slit wall.

1). Color: Orange for multi-mode and yellow for single-mode. White for both in plenum areas.

2). Sizes, as required, shall be:

a). 3/4 inch inside diameter. b).
1 inch inside diameter.
c). 1 1/4 inch inside diameter.

F. If deviation from color specified above is required, submit request for approval to A/E and Owner. Chose from the following inner-duct colors black, orange, gray, blue, brown, red, white, green, and yellow. No two inner-duct of the same size in a conduit are to be of the same color.

29.2 INNER-DUCT PLUGS and COUPLINGS

A. Acceptable Manufacturers:

1). Duriline.

2). Carlon.

B. Couplings: metallic or nonmetallic quick connect, reverse threaded, and Schedule 40 couplings for connecting sections of installed inner-duct. Inner- duct couplings shall be compatible with and intended for the purpose of coupling inner-duct installed.

C. Inner-duct plugs shall be compression-type inner-duct plug for sealing inner- duct, with an eyebolt for securing the pull tape. The plug shall be supplied with a wing nut for hand tightening. These plugs shall be used in wet locations to keep water out of unused inner-duct:

1). Sizes:

a). 3/4-inch blank duct plug. b).
1 inch blank duct plug.
c). 1 1/4 inch blank duct plug.

D. Inner-duct caps shall be removable push-in caps for plugging inner-duct, with molded-in eye for securing of pull tape. These plugs shall be used in dry locations to keep vermin out of unused inner-duct:

1). Sizes:

a). 3/4-inch blank duct plug. b).
1 inch blank duct plug.
c). 1 1/4 inch blank duct plug.

29.3 PULL TAPE

A. Acceptable Manufacturers:

1). Duriline.

2). Carlon.

B. Pull Tape shall be measuring and pulling tape constructed of synthetic fiber with plastic jacket, printed with accurate sequential footage marks.

29.4 PENETRATION SEALING SYSTEMS (WATER)

A. Acceptable Manufacturers:

1). Duriline.

2). Carlon.

B. Duct water seal shall be manufactured for the intended use to seal underground and entrance duct openings where cable is installed.

EXECUTION

30.1 PREPARATION

A. Survey the work area for conflicts or hazardous conditions that may cause damage to the cabling system or harm to personnel or property in the construction area.

B. Ensure that conduit raceway ends are capped with conduit collars to protect cables. Be sure raceway, cable tray routes pull boxes and outlet boxes are free from debris, obstructions or sharp edges.

C. Inspect firestopping installation by others between building structure and/or inner-duct and conduit, sleeves, wire way, and cable tray to verify integrity of installation. Advise the Contractor and Architect of the need to remove the firestopping material and submit a written record of the locations. Remove and store the firestopping material in a safe location. Reinstall the firestopping material after the cable installation is complete in accordance with the provisions of Specification 07270 - Firestopping.

D. Remove and replace ceiling tile as required to obtain access for completing the subcontractor's portion of the project. Repair any ceiling tile damage caused by this activity.

E. Protect adjacent surfaces from damage during water duct seal or firestop installation; repair any damage.

30.2 APPLICATION

A. Under-slab and Underground Raceways: Outdoor inner-duct.

B. Aboveground Raceways, Exterior and Interior: Indoor inner-duct.

C. Interior Exposed Locations, Including Cable Tray Installations: 1).

Non-plenum Areas: Riser rated inner-duct.

2). Plenum Areas: Plenum listed inner-duct.

D. Cable Tray to Active Equipment Mounting Transitions as indicated on Project Drawings: Split-corrugated inner-duct.

30.3 INNER-DUCT INSTALLATION

A. Pull inner-duct through conduit and wire ways, or place inner-duct in cable trays using continuous unspliced lengths of inner-duct between manholes, pull boxes, and/or section termination points.

B. Conduit designated to have inner-duct shall contain the following quantities of inner-duct:

1). 4 inch conduits shall contain the following quantity, size, and color of inner-duct:

a). One 1 1/4 inch yellow or white. b).
Two 1-inch blue and orange.
c). Three 3/4-inch red, black, and gray.

2). 3 inch conduits shall contain the following quantity, size, and color of inner-duct:

a). One 1 1/4 inch yellow or white. b).
One 1-inch blue.
c). One or two 3/4-inch red and black (number dependent on installer's recommendation).

3). 2 1/2 inch conduits shall contain the following quantity, size, and color of inner-duct:

a). Two 3/4-inch red and black (number dependent on installer's recommendation). Inner-duct fill ratios shall be calculated off schedule 40 PVC.

4). 2 inch conduits and smaller shall not require inner-duct.

C. Cut inner-duct square; deburr cut ends.

D. Bring inner-duct to the shoulder of fittings and couplings and fasten securely.

E. Wipe inner-duct and fittings clean and dry before joining. Apply full, even coat of cement to entire area that will be inserted into fitting. Let joint cure for a minimum of 20 minutes.

F. Provide suitable inner-duct slack in manholes, handholes, pull boxes, and at turns to ensure that there is no kinking or binding of the tubing.

G. Make changes in direction of communications inner-duct runs with sweeps of the longest possible radius and a minimum of 10 times the inside diameter of the inner-duct.

H. Inner-duct Mountings, Hangers, and Attachments: When exposed indoors or in manholes, hold inner-duct firmly in place using independent support.

I. Underground Conduit Installation: After inner-duct is installed in conduit entering structures, seal openings inside conduit around inner-duct at first box or outlet to prevent the entrance of gases, liquids, or rodents into the structure. Use duct water seal.

J. Manhole and Handhole Installation:

    1). At manholes serving as a junction location, pull the exposed end of the inner-duct to the far end of the vault, install manufactured duct plugs, and secure to cable racks.

    2). At other locations where the ends of inner-duct sections appear in a manhole or handhole, join the pull tape, and then splice inner-duct sections together using watertight couplings which do not reduce the inside diameter of the inner-duct.

K. Indoor Conduit Installation:

    1). Arrange inner-duct neatly and remove surplus length. Provide trained and bundled inner-duct pigtails extending at least 18 inches beyond exposed conduit openings.

    2). At locations where the ends of inner-duct sections appear in a pull box, join the pull tape, and then splice inner-duct sections together using couplings which do not reduce the inside diameter of the inner-duct.

    3). Inner-duct shall not be continuous through J-Boxes used for 90-degree bends. This is the least preferred method for turning a corner and as such its use is subject to OWNERS APPROVAL in each incident.

    4). Cable Tray Installation: Tie wrap inner-duct to one side of vertical ladder rack every 3 feet minimum, and to one side of horizontal ladder- type cable tray every 6 feet minimum.

L. Following the installation the contractor shall visually inspect inner-duct, remove burrs at openings, and, clean inner-duct interior of debris.

30.4 PULL TAPE AND DUCT PLUG INSTALLATION

A. Inner-duct 1-inch and larger shall be ordered with a pull tape (mule-tape) preinstalled. Following inner-duct installation tie the pull tape securely to duct- plug at each location.

B. Following 3/4–inch inner-duct installation; install a pull tape (mule-tape) with preprinted foot markers in inner-duct sections. Tie the pull tape securely to duct-plug at each location.

C. Verify lengths at the time of installation and provide record and submit documentation.

D. Following inner-duct and pull tape installation, cap or plug inner-duct with manufactured seals to prevent moisture or foreign matter from entering until cable installation starts.

E. Immediately after installation, install removable plugs in empty conduit and wire way openings. For underground and wet area inner-duct openings, use screw- tight, removable, watertight, and dust-tight duct plugs.

30.5 PENETRATIONS

A. Seal conduits and inner-duct entering structures at the first box or outlet to prevent entrance into the structure of gases, liquids, or rodents. Conduit with inner-duct installed shall be sealed with duct water seal between conduit and inner-duct.

B. Inner-duct with cable installed shall be sealed with appreciate sized single cable duct-plug or duct water seal between inner-duct and cables.

C. Raceway Penetrations shall be sealed between inner-duct and conduit or wire way at exposed penetration locations.

D. Cable Tray Penetrations: The subcontractor shall remove and protect firestopping materials throughout the construction period in a clean and properly protected condition to maintain each assembly without any indication of damage. After inner-duct installation, restore fire barrier penetration sealing materials to provide required protection.

30.6 PROTECTION

A. Protect products from the effects of moisture, corrosion, and physical damage during construction. Except during installation activity in a section, keep openings in conduit, wire way, and inner-duct capped with manufactured seals.

PART 11 COMMUNICATIONS CABLE TRAYS

GENERAL

31.1 SUMMARY

A. Section Includes: This Section specifies the requirements necessary to furnish and install communications cable trays including:

1). VE 1, Section 4, Welded Steel Wire Mesh metallic cable tray systems. 2).

VE 1 cable tray banding and warning signage.

3). Wire Basket Tray.

B. Related Specifications: This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for communications cable trays:

1). Specification 07270 - Firestopping.

31.2 SUMBITTALS

A. Provide the following information in addition to the standard requirements with the Bid:

1). For cable tray, show location of maximum simple beam deflection of tray for loading specified.

2). Layout drawings indicating cable tray type, dimensions, support points, elevation change methods, etc. Include a list of proposed fittings and accessories.

31.3 CLOSEOUT SUBMITTALS

A. Project Record Documents: B.
1). Plan Drawings showing completions and as-built corrections that indicate type, size, placement, routing, and/or length for cable tray components.

PRODUCTS

32.1 GENERAL

A. Cable Tray Materials and Equipment: Labeled and/or listed as acceptable to the authority having jurisdiction as suitable for the use intended.

32.2 METALLIC CABLE TRAY SYSTEMS

A. VE 1 Tray Systems made of straight sections, fittings, and accessories as defined in NEMA standards publication No. VE 1, which are UL-classified as equipment grounding conductors:

1). Straight-Section Type: Welded Steel Wire Mesh-type cable tray as shown on the Drawings.

2). Span Class Designation: Welded Steel Wire Mesh –type 8 feet as shown on Project Drawings.

3). Load Class Designation: Class A (50 pounds per foot) as shown on Project Drawings.

4). Inside Depth: Shall be as shown on the Drawings. Outside depths shall not exceed inside depths by more than 1 1/4 inches.

5). Inside Width: Shall be as shown on the Drawings. Outside widths shall not exceed inside widths by more than 4 inches.

6). Covers: Flanged, solid cover attached with standard cover clamps when indicated on Project Drawings.

7). Accessories: Manufacturer’s standard clamps, hangers, brackets, splice plates and hardware, label clips, wall mount attachments, reducer plates, blind ends, ladder and trough dropouts, barrier strips, connectors, conduit and box supports, grounding lugs and grounding straps and seismic bracing kits.

B. Wire Basket Tray

1). Wire basket support systems are defined to include, but are not limited to straight sections of continuous wire mesh, field formed horizontal and vertical bends, tees, drop outs, supports, and accessories.

2). Material construction: high strength steel wires formed into a standard 2 inch by 4 inch wire mesh pattern with intersecting wires welded together. Wire ends along wire basket sides (flanges) shall be rounded during manufacturing for safety of cables and installers.

3). Fittings and accessories: field formed as needed, provide manufacturer standard fasteners, hardware, and other items required to complete the Installation as indicated on the Drawings.

4). Width: As indicated on project drawings. 5).

Depth: As indicated on project drawings.

32.3 Cable Tray Warning Signage and Banding

A. Warning Signage: 1/2 inch-high black letters on yellow plastic nameplate with the following wording: WARNING! DO NOT USE CABLE TRAY AS WALKWAY, LADDER, OR SUPPORT. USE ONLY AS MECHANICAL SUPPORT FOR CABLES!

B. Fire Barrier (Wall/Floor) Warning Signage: 1/2 inch-high white letters on red plastic nameplate with the following wording to be secured on the tray at each side of the fire barrier to read: WARNING! FIRE BARRIER – PROVIDE FIRE BLOCK TO MAINTAIN BARRIER INTEGRITY!

C. Cable Tray Banding: 3-inch wide yellow tape. As per "Communication Cable Trays" section.

EXECUTION

33.1 APPLICATION

A. Install the following types of cable tray in the locations listed and as indicated on Project Drawings:

1). Interior Pathways Extending Beyond Communication Rooms:

1. Other Areas: Welded Steel Wire Mesh–Type where vertical clearance is only 4 to 8 inches, or where elevation and directional changes make Ladder-Type cable tray impractical. As shown on project drawings.

2). Communications Room Pathways:

a). Welded Steel Wire Mesh–Type at low level at an elevation just above the equipment racks or as required per project drawings.

33.2 GENERAL INSTALLATION

A. Cut standard straight sections of materials to length in the field.

B. Deburr and file rough cable tray edges and any cut sections.

C. Cable tray locations shown on the Drawings are approximate unless dimensioned.

D. Install cable tray as shown on the Drawings and securely attach under the provisions of the “Communications Supporting Methods” section and label under the provisions of the “Communications Identification” section.

E. All cable tray shall be accessible.

F. Maintain minimum 8 inch clearance between cable tray and piping. Locate cable tray at least 12 inches away from heat sources such as parallel runs of flues, steam or hot water pipes, and heating appliances.

G. Run exposed and concealed cable tray parallel or perpendicular to walls, structural members, or intersections of vertical planes to maintain headroom and provide a neat appearance.

H. Passageways shall not be obstructed.

I. Cable tray shall be routed within the assigned communications utility space.

J. Install appropriate cable tray bends, dropouts, and other accessories to protect minimum cable bend radius and provide adequate support at locations where cable direction changes occur.

33.3 CABLE TRAY INSTALLATION

A. Installation: In conformance with NEMA VE 1 requirements.

B. Install cable tray support at each connection point, at the end of each run, and at other points to maintain spacing between supports of 8 feet maximum or as per manufacturer load/support spacing chart recommends. Trays shall be level with respect to grade plus or minus 1/8 inch per 10 feet, or 1/2 inch cumulative exclusive of required elevation changes as shown on Project Drawings.

C. Use expansion connectors where indicated on project drawings or where cable tray crosses building expansion or seismic joints.

D. Elevation Change Methods:

1). Install standard vertical outside bends, and inside bends with a 12 inch or longer bend radius, using connecting straight sections as required.

2). Join 1, 2, or 3 straight section segments using vertical adjustable splice plates, as required, to maintain vertical changes at each adjustable side plate to less than 30 degrees.

3). The spacing between cable support surfaces and/or ladder rungs at elevation change locations shall not exceed 1/2 inch.

4). Cable and/or Innerduct shall be affixed to the cable tray at all direction and elevation changes.

5). Install cable tray rollers at the lower end of all upwards elevations changes, to prevent cable from coming out of tray during installation.

E. When Project Drawings call for the installation of cable tray dropouts, provide plastic grommets around openings to prevent cable abrasion.

33.4 PENETRATIONS

A. Provide fire stopping at fire barrier penetrations as specified in Specification 07270 - Firestopping.

33.5 CABLE TRAY BANDING AND SIGNAGE

A. Provide as per the “Communications Identification” section.

B. Provide warning signs on alternating sides at 50 foot centers along route of cable tray located to be visible. Install fire barrier warning signs at fire barrier penetrations.

33.6 PROTECTION

A. Following installation, protect products from the effects of moisture, corrosion, and physical damage during construction.

PART 12 COMMUNICATIONS HANDHOLES AND MANHOLES

GENERAL

34.1 SUMMARY

A. Section Includes: This Section specifies the requirements necessary to furnish and install communications handholes and manholes including:

1). Prefabricated concrete handholes. 2).

Prefabricated concrete manholes. 3).

Manhole accessories.

B. Related Specification: This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for communications handholes and manholes:

1). Section 02200 - Earthwork.

34.1 SUBMITTALS

A. Provide the following information in addition to the standard requirements with the Bid:

1). Material specifications showing external and internal dimensions of handholes and manholes, and location of fittings, accessories, conduit terminators, etc.

PRODUCTS

35.1 BASE STOCK

A. Clean 1 inch gravel or crushed rock uniformly graded from coarse to fine with less than 8 percent by weight passing a No. 200 sieve.

35.2 COMMUNICATIONS HANDHOLE

A. Precast Concrete Handholes for Underground Installations:

1). Construction: Rectangular vault with cable entrance at center bottom of each end.

2). Precast Concrete: 4,500 psi strength at 28 days.

3). Reinforcing: AASHTO H 20, bridge loading.

4). Handhole Dimensions: As indicated on the drawings.

5). Channel struts embedded in each interior side, as indicated on project drawings.

6). One 1/4 inch galvanized diamond plate steel covers, with bolt locks, hinged to an adjustable frame as shown on the drawings.

7). Grounding: Provide ground bonding ribbon mounted on the interior wall surface that is bonded to precast reinforcing bar structure.

35.3 COMMUNCATIONS MANHOLE

A. Precast Concrete Manholes for Underground Installations:

1). Construction: In modular sections with tongue and groove joints. 2).

Precast Concrete: 4,500 psi strength at 28 days.

3). Reinforcing: AASHTO H 20, bridge loading.

4). Manhole Dimensions: As indicated on the drawings.

5). Core drill and provide watertight duct connections for 4 inch ID underground utility conduits as indicated on project drawings.

6). Include 12 inch x 12 inch drain sump in base section.

7). Include cable pulling irons at each corner and an appropriate number of pulling insert anchors.

8). Include 2 or more horizontal rows of “C” channel inserts for cable racking, as indicated on project drawings.

9). Grounding: Provide ground bonding ribbon mounted on the interior wall surface that is bonded to precast reinforcing bar structure.

10). Telecommunication grounding shall be per the “Communications Cross Connections” section.

11). Core drill and seal hole for grounding rod insertion in bottom of manhole.

B. Manhole Frames and Covers:

1). Material: Machined cast iron with lift slots and bolt down holes. 2).

Cover Diameter: 30 inches.

3). Cover Loading: AASHTO H 20.

4). Cover Designation: The word COMM shall be provided on welded plate on the cover.

5). Grade Rings: 4 inch and 6 inch thick rings to fit manhole opening.

35.4 MANHOLE ACCESSORIES

A. Pulling Eyes: 1/2 inch inserts with 1 1/4 inch eyebolts sufficient to provide floor and ceiling pulling support opposite each cable entry location.

B. Cable Rack Supports: 1 inch by 1/2 inch S and L cable rack supports as required to mount racking. Hot dipped galvanized steel with mounting holes for 1/2 inch bolts.

C. Cable Racking: Hot dipped galvanized steel cable racking with cable arm mounting slots. Mount cable racking vertically at 2 feet on center.

D. Ground Rods: Copper or copper clad steel, 5/8 inch in diameter, 10 feet long, driven full length into the earth, through bottom of manhole.

EXECUTION

36.1 PREPARATION

A. Excavate, install base material, and compact base material in accordance with Specification 02200, - Earthwork.

1). Remove water from excavation. Place a minimum of 12 inches of base rock at bottom and sides and thoroughly compact with mechanical vibrating or power tamper.

2). If material in bottom of trench is unsuitable for supporting manhole, excavate as directed by Contractor. Backfill to required grade using 3/4 inch-maximum size rock.

36.2 INSTALLATION

A. Precast Concrete Handholes:

1). Install precast vault and cover frame in accordance with manufacturer’s instructions.

2). Install handholes plumb.

3). Set top on each handhole to finished elevation as indicated.

B. Precast Concrete Manholes:

1). Carefully inspect manhole sections to be joined. Do not use sections with chips or cracks.

2). Install and seal precast sections using preformed plastic gaskets in accordance with manufacturer’s instructions.

3). Install manholes plumb and set each manhole to finished elevation as indicated on the drawings.

4). Backfill around manholes using the same backfill specified for the adjacent duct bank.

5). Waterproof exterior surfaces, joints, and interruptions of manholes after installation.

6). Use cast iron frame and grade adjusting ring sections to bring the manhole entrance cover to proper elevation.

a). Install grade rings, frames, and covers on top of manholes to positively prevent infiltration of surface or groundwater into manholes.
b). Install grade rings in conformance with the details shown on the drawings and to the height required to match final grade. Lay grade rings in mortar with sides plumb and tops level. Seal joints with mortar as required. Extensions shall be watertight.
c). Set frames so tops of covers are 1/2-inch above surface of adjoining pavement or ground surface unless otherwise shown or directed.

C. Manhole Accessories:

1). Where indicated, install drains in manholes and terminate in crushed gravel bed.

2). Install a ground rod inside manholes. Connect ground rods to the bonding ribbons inside the manhole with a No. 6 AWG copper conductor.

3). Install pulling eyes and attach cable racks to “C” channel inserts.

D. Manhole And Handhole Identification:

1). Manholes and handholes shall be identified as per the “Communications Identification” section.

2). Field-stamp covers with manhole numbers as indicated on the drawings, e.g., TH001.

PART 13 INTRA-BUILDING BACKBONE

GENERAL

37.1 SUMMARY

A. This Section specifies the requirements to furnish and install intra-building backbone unshielded twisted-pair (UTP) and fiber optic cable distribution subsystem, including:

1). UTP cable placement and termination.

2). Fiber optic cable placement and termination.

37.2 SYSTEM DESCRIPTION

A. This backbone distribution subsystem refers to intra-building twisted-pair and fiber optic communications cabling interconnecting utility entrance facilities, Building Communications Rooms (BCRs), Satellite Communications Rooms (SCRs), and Satellite Communications Cabinets (SCCs).

B. There shall not be any splices in the Intra-Building Backbone Communications Distribution Subsystem.

37.3 CLOSEOUT SUBMITTALS

A. Project Record Documents

1). Submit backbone cable schedule check-off list for each installed, terminated, and labeled cable showing completions and as built corrections.

2). Submit test results and verification for each cable pair and fiber optic strand installed as are outlined in the “General Communications Requirements” section and the “Communications Cable Testing Methods” section.

3). Provide wiring block and fiber distribution unit (FDU) addressing and/or numbering lists for cables installed.

PRODUCTS

38.1 MULTI-PAIR UNSHIELDED TWISTED-PAIR BACKBONE CABLE

A. Acceptable Manufacturers:

1). Berk-Tek.

B. Cable Types:

1). Refer to the “Communications Cable and Accessories” section for cable type specifications.

2). As a minimum, a 400 pair ARMM Riser Cable shall be run between the BCR and Each SCR.

3). As a minimum, a 100 pair ARMM Riser Cable shall be run between the BCR and each SCC.

4). Other pair counts and/or cable configurations may be used when approved by the Owner and shown on Project Drawings.

38.2 MULTIMODE FIBER OPTIC CABLE

A. Acceptable Manufacturers:

1). Corning.

2). Berk-Tek.

B. Cable Types:

1). Refer to Section the “Communications Cable and Accessories” section for cable type specifications.

2). As a minimum, a 36-fiber count Riser Cable shall be run between the BCR and each SCR.

3). As a minimum, a 12-fiber count Riser Cable shall be run between the BCR and each SCC.

4). Other fiber counts and/or cable configurations may be used when approved by the Owner and shown on Project Drawings.

38.3 SINGLE-MODE FIBER OPTIC CABLE

A. Acceptable Manufacturers:

1). Corning.

2). Berk-Tek.

B. Cable Types:

1). Refer to Specification 17100 – Communications Cable and Accessories for cable type specifications.

2). When required for specific applications, as a minimum, a 12-fiber count Riser Cable shall be run between the BCR and each SCR.

3). When required or specific applications, as a minimum, a 6-fiber count Riser Cable shall be run between the BCR and each SCC.

4). Other fiber counts and/or cable configurations may be used when approved by the Owner and shown on Project Drawings.

38.4 COMPOSITE FIBER OPTIC CABLE

A. Acceptable Manufacturers:

1). Corning.

2). Berk-Tek.

B. Cable Types:

1). Refer to the “Communications Cable and Accessories” section.

2). When required for specific applications, as a minimum, a 36-fiber count multimode and 12-fiber count single-mode Riser Cable shall be run between the BCR and each SCR.

3). When required for specific applications, as a minimum, a 12-fiber count multimode and 6 fiber count single-mode Riser Cable shall be run between the BCR and each SCR.

4). Other pair counts and/or cable configurations may be used when approved by the Owner and shown on the Project Drawings.

38.5 DUCT PLUGS

A. Acceptable Manufacturers:

1). Carlon

2). PNA

3). Duraline.

B. All empty conduit that runs between the BCR and each SCR or Sc shall be plugged with proper Conduit Plug. Refer to “Communications Raceways and Boxes” section for proper application.

C. All empty inner-duct that run between the BCR and each SCR or SCC shall be plugged with proper Inner-Duct Plug. Refer to the “Communications Inner- Duct” section for proper application.

EXECUTION

39.1 PREPARATION

A. Survey the work area for conflicts or hazardous conditions that may cause damage to the cabling system or harm to personnel or property in the construction area.

B. Ensure that conduit raceway ends are capped with conduit collars to protect cables. Be sure raceway, cable tray routes pull boxes, and outlet boxes are free from debris, obstructions, or sharp edges.

C. Inspect firestopping installation by others between building structure and/or inner-duct and conduit, sleeves, wireway, and cable tray to verify integrity of installation. Advise the Architect of the need to remove the firestopping material and submit a written record of the locations. Remove and store the firestopping material in a safe location. Reinstall the firestopping material after the cable installation is complete in accordance with the provisions of Specification 07270
- Firestopping.

D. Remove and replace ceiling tile as required to obtain access for completing the Contractor's portion of the project. Repair any ceiling tile damage caused by this activity.

E. Protect adjacent surfaces from damage during water seal or firestop installation; repair any damage.

F. Conduct fiber optic cable reel tests as specified in the “Communications Cable Testing Methods” section.

39.2 INSTALLATION

A. Provide and install inner-duct in accordance with the provisions of the “Communications Inner-Duct” section.

B. Place and terminate twisted-pair and fiber optic cable in accordance with the provisions of the “Communications Cable and Accessories” section.

C. Underground Conduit Openings: Seal the high end of underground duct section openings to protect against water and gas leaks; use water seal with installed cables and removable 4 inch screw-tight duct plugs in vacant underground duct openings.

D. Raceway Entrances: Seal raceways entering structures, including conduit and innerduct with cable installed, at the first box or outlet to prevent the entrance of gases, liquids, or rodents into the structure.

1). Inspect entrance seal installation by others between building structure and/or innerduct and conduit to verify integrity of installation.

2). Empty Conduit: Unless provided by others, install removable screw- tight duct plugs.

3). Conduit with Innerduct Installed: Install suitable duct water seal between conduit and innerduct.

4). After cable is installed, install suitable duct water seal between conduit and cable or between innerduct and cable.

E. Furnish and install wire and hardware required to properly ground and bond communications cable and equipment in accordance with the provisions of the “Communications Grounding” section. The following additional requirements apply.

1). Serve each metallic splice closure by a minimum of a 6 AWG insulated copper wire connected to a building ground source.

2). Bond metallic cable sheaths in multi-pair communications cables together at each splicing and/or terminating location to provide 100 percent metallic sheath continuity throughout the communications distribution system.

a). At termination points, install a cable shield bonding connector to provide a screw stud connection for ground wire. Use a bonding jumper to connect the cable shield connector to an appropriate ground source like an equipment rack or cabinet ground bar.
b). Bond metallic cable shields together within splice closures using cable shield bonding connectors or the splice case grounding and bonding accessories provided by the splice case manufacturer.

3). If a metallic shield is not provided within the sheath around a multi-pair cable core, install a green, insulated 6 AWG stranded ground conductor run parallel to the multi-pair cable.

F. Identify and label the system in accordance with the provisions of the "Communications Identification" section.

39.3 FINAL TEST

A. Conduct final system testing in accordance with the provisions of the "Communications Cable Testing Methods" section.

39.4 PROTECTION

A. Protect adjacent surfaces from damage during water seal or firestop installation; repair any damage.

PART 14 HORIZONTAL COMMUNICATIONS DISTRIBUTION SUBSYSTEM

GENERAL

40.1 SUMMARY

A. Section Includes:

1). This Section specifies the requirements necessary to furnish and install a horizontal unshielded twisted-pair (UTP) distribution subsystem, including:

a). UTP cable placement and termination.
b). Telecommunications Outlet (T.O.) components.

2). This Section specifies the requirements necessary to furnish and install a horizontal fiber optic distribution subsystem, including:

a). Fiber optic cable placement and termination. b). Telecommunications Outlet (T.O.) components.

B. Related Specification

1). Related Work: This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for the horizontal communications distribution subsystem:

a). Specification 07270 - Firestopping.

40.2 SYSTEM DESCRIPTION

A. The Horizontal Communications Distribution Subsystem refers to intra building UTP, and fiber optic communications cabling connecting Building Communications Rooms (BCR), Satellite Communications Rooms (SCRs), and Satellite Communications Cabinets (SCCs), to T.O.s located at the work area.

B. Horizontal cables shall be routed from the cross connect field in the BCR, SCR, or SCC, to the work area in cable tray, wireway, and/or conduits defined as communication pathway per drawings. In spaces below raised floors and above suspended ceiling tile, horizontal cable support is provided by cable tray and/or conduits to general work area locations.

C. Horizontal cables shall be terminated as per Horizontal Cable Schedules.

40.1 CLOSEOUT SUBMITTALS

A. Project Record Documents

1). Submit horizontal cable schedule check off list for each installed, terminated, labeled, and tested cable showing completions and as built corrections.
2).
3). Provide wiring block addressing and/or numbering lists for cables installed.

PRODUCTS

41.1 GENERAL

A. Labeling: Refer to "Communications Identification" section.

B. Cable: Refer to "Communications Cable and Accessories" section.

C. Cable Suspension System: Refer to "Communications Supporting Methods".

41.2 TELECOMMUNICATIONS OUTLETS

A. Acceptable Manufacturers:

1) CAT6A Jack Module: Leviton Network Solutions part # 6AUJK-SV6
2) Outlet Faceplate: Leviton Network Solutions part # 42081-4WS
3) Wall Phone Faceplate: Leviton Network Solutions part #41084-1SP
4) Surface Mounted Outlet Box: Leviton Network Solutions part # 4S089- 2WP
5) Faceplate Blank Insert: Leviton Network Solutions part #41084-0BW

B. Wall Phone Faceplate: Single port wall mount outlet used as an interface for plug in modular wall telephone sets. Stainless steel material with 2 metal studs located to be equal to the mounting slots of a modular wall phone set. The wall plate shall have the capability to be mounted with 2 screws to a standard single gang mud-ring on a 4-square outlet box. The port shall be 6P6C RJ11 USOC connector with punchdown termination.

C. Telecommunications Outlet System:

1). Multi-port wall mount outlet faceplate used as an interface for Voice, Data, Video or Fiber Optic Networks.

a). Faceplate: Plastic material with 1 through 12 outlet port capability.

1. The single gang 1 through 6 outlet wall plate shall have the capability to be mounted with 2 screws to a standard single gang mud-ring on a 5 square outlet box

2. The double-gang 1 through 12 outlet wall plate shall have the capability to be mounted with 4 screws to a standard double- gang outlet box.

b). Faceplate shall be capable of allowing outlet port modules to be interchanged to allow for Category 6A ports and/or SC style fiber optic couplings.

2). Surface Mounted Multi port multi media modular outlet box used as an interface for Voice, Data, Video, or Fiber Optic Networks.

a). Single Module Box: Plastic material with 2 outlet port capability. The box shall be screw mounted to system furniture or in equipment cabinets. They cannot be mounted on outlet boxes nor be used for fiber optic modules.
b). Dual Module Box: Plastic material with 1 through 4 outlet port capability. The box shall to be mounted with 2 screws to walls, system furniture or in equipment cabinets.
c). Three Module Box: Plastic material with 1 through 6 outlet port capability. The box shall have the capability to be mounted with 2 screws to a standard 4-inch square or 2 gang outlet box with a single gang mud ring, walls, system furniture or in equipment cabinets. This box shall be the primary method for providing fiber optic modules when a surface mounted box is required. This box shall for 3 fiber optic modules when mounted on an outlet box. This box shall have built-in fiber optic cable management for proper fiber cable bend radius control and storage.
d). Four Module Box: Plastic material with 1 through 8 outlet port capability. The box shall have the capability to be mounted with 2 screws to a standard 4-inch square or 2 gang outlet box with a single gang mud ring, walls, system furniture or in equipment cabinets. This box shall for 2 fiber optic modules when mounted on an outlet box.
e). These boxes shall be capable of allowing outlet port modules to be interchanged to allow for Category 6A ports and/or SC style fiber optic couplings.

3). The UTP outlet port modules shall be a single RJ45 Category 6A compliant with a 568B configuration and tool-free termination. The UTP outlet port module shall be of the same manufacturer and intended for use with the faceplates and boxes above.

4). The fiber optic outlet port module shall be a recessed single or dual angled duplex SC of the same manufacturer and intended for use with the faceplates and boxes above.

5). Accessories:

a). In any situation where an outlet port position in the outlet frame is not occupied by an outlet port module a blank insert is to be supplied and installed. The blank insert module shall be of the same manufacturer and intended for use with the outlet frame. The blank module shall come in .5, 1 and 1.5 module heights.
b). Single Gang Faceplate Extension: Plastic plate required for use in fiber optic applications. These required extensions provide the extra space for proper fiber bend radius and storage.
c). Video F Connector modules in single or dual configuration.
d). Spring loaded shutter to cover RJ45 jack openings to project from dust mist or other contamination.
e). Module designation tabs that snap into module face to provide identification of each jack port.

<u>EXECUTION</u>

42.1 PREPARATION

A. Survey the work area for conflicts or hazardous conditions that may cause damage to the cabling system or harm to personnel or property in the construction area.

B. Ensure that conduit raceway ends are capped with conduit collars to protect cables. Be sure raceway, cable tray routes pull boxes, and outlet boxes are free from debris, obstructions, or sharp edges.

C. Inspect firestopping installation by others between building structure and/or innerduct and conduit, sleeves, wireway, and cable tray to verify integrity of installation. Advise the Contractor and Architect of the need to remove the firestopping material and submit a written record of the locations. Remove and store the firestopping material in a safe location. Reinstall the firestopping material after the cable installation is complete in accordance with the provisions of Specification 07270 - Firestopping.

D. Unless excluded from work install conduit sleeves with insulating throat bushings or fire barrier sealing systems in openings where open cable passes through fire rated walls and floors. Install the firestopping material after the cable installation is complete in accordance with the provisions of Specification 07270 - Firestopping.

E. Remove and replace ceiling tile as required to obtain access for completing the Subcontractor's portion of the project. Repair or replace any ceiling tile damage caused by this activity.

42.2 INSTALLATION

A. Place UTP and/or fiber optic cable in accordance with the provisions of the “Communications Cable and Accessories” section.

B. Horizontal Cable Support: Where specified on the Drawings between T.O.s and raceway or cable tray entrances, support horizontal cables from structure using Cable Suspension System in accordance with the provision to the “Communications Supporting Methods” section. Cable Suspension System, when used, must be dedicated to communications cable support. There cannot be any attachment or contact with similar wires used for ceiling grid support.

C. Modular Furniture/Power Poles: At the entry point to power poles and modular furniture partitions, tie or enclose the cable bundles in a manner that does not restrict the bend radius of the components.

D. Hazardous Areas: Ensure that each conduit running from a non hazardous area to a hazardous area and each conduit entering an enclosure within a hazardous area is provided with a sealing fitting that meets applicable NEC 500 requirements. After installing and terminating cables serving hazardous locations, seal cables in conduit by filling sealing fittings using an approved sealing compound of the same manufacture.

E. Care shall be taken to assure that cables installed in outlet boxes are not kinked, pinched or cut when the T.O. wall plates is installed.

F. Cables shall be attached to the T.O. wall plate as per the T.O. wall plates manufacturer's recommended methods.

G. Terminate UTP and/or fiber optic cable in accordance with the Horizontal Cable Schedules and the provisions of the “Communications Cable and Accessories” section.

H. General Location of T.O.s shall be as shown on drawings. Conflicts that inhibit the installation or use of the T.O. shall be recorded and reported to the Architect, Contractor and Owner.

I. Flush to the wall mounted T.O. wall plates shall be attached to the outlet box provided for that purpose. The wall plate shall be aligned parallel and perpendicular to the building lines. Record and report to the Contractor and Architect T.O. locations where the T.O. wall plate does not cover the opening in the wall or where the outlet box installation will not allow the T.O. wall plate to be installed parallel and perpendicular to the building lines.

J. Surface-mounted T.O. wall plates shall be attached to the outlet box provided for that purpose. The wall plate shall be aligned parallel and perpendicular to the building lines. Record and report to the Contractor and Architect T.O. locations where the outlet box installation will not allow the T.O. wall plate to be installed parallel and perpendicular to the building lines.

K. Furniture mounted T.O.s shall be securely attached to furniture with fasteners designed and provided for that purpose. The communication contractor shall coordinate with the furniture supplier to determine exact T.O. location and mounting methods. The T.O. shall be aligned parallel and perpendicular to the furniture lines.

L. T.O.s mounted beneath a raised floor shall be securely attached to the raised floor pedestal. The communication contractor shall coordinate with the raised floor supplier to determine exact size and type of fasteners for the mounting T.O.s to the pedestal.

M. T.O.s mounted inside cabinets not supplied by the communication contractor. The communication contractor shall coordinate with system owner of the cabinet to determine exact location and type of fasteners required for the mounting T.O.s in the cabinet.

N. Identify and label the cable and T.O.s in accordance with the provisions of the "Communications Identification" section.

O. Place blank covers over spare T.O. and T.O. jack openings.

42. FINAL TEST

A. Conduct final system testing in accordance with the provisions of the "Communications System Testing" section.

42.4 PROTECTION

A. Protect adjacent surfaces from damage during water seal or firestop installation; repair any damage.

END OF SPECIFICATION

SECTION 16780

CCTV SYSTEM SPECIFICATION

PART 1 GENERAL

1.1 SUMMARY

A. In general, the CCTV Contractor selected shall be responsible for providing a complete and operable CCTV system as described herein. This work shall include final connections to existing CCTV equipment and successful operation of the new equipment with the existing equipment. The CCTV Contractor shall provide a written response to each paragraph in this Specification to verify compliance.

1.2 RELATED DOCUMENTS

1.3 SUBMITTALS

A. Before executing any work, submit for approval, 7 copies of information (in completely marked and coordinated package) to assure full compliance with the contract requirements. Information shall be submitted for all material and equipment the CCTV Contractor proposes to furnish to accomplish the contract work. Submittals for each manufactured item shall contain descriptive literature (equipment specification), equipment drawings, diagrams and catalog cuts of the proposed equipment. The submittal shall also include manufacturer names, catalog models and numbers, layout dimensions and capacities. Drawings shall indicate adequate clearance for operation, maintenance and replacement of operating equipment and devices.

B. The following operational and maintenance material shall be submitted at the completion of the work:

1). Three bound books, with index and index tabs, to include the following for all products and devices furnished under thisSpecification.

a). Hardware manuals
b). Maintenance manuals

C. List of recommended spare equipment, along with quantities the Owner should maintain on site.

D. Final approved set of all shop drawingsubmittals.

1.4 INSTALLATION AND START-UP REQUIREMENTS

A. Micron will specify locations of all field devices prior to installation by the CCTV Contractor and will be shown on EngineeringDesign package documents.

B. Provide a written punch list of items remaining to be completed to Micron and/or General Contractor two (2) weeks prior to startup date.

1.5 Start-up and testing of the system will be done by the CCTV Contractor with the assistance of Micron. During start-up the CCTV Contractor will be responsible for repairing any mistakes in wiring or installation immediately. As a minimum, one CCTV technician familiar with the installation shall be available at all times during start-up for the repairs.

1.6 AS-BUILT DOCUMENTATION

A. The Contractor shall keep one print of each drawing at their shop and in the field. These prints are to be marked-up as work progresses. The Contractor will provide as-built drawings at the end of this job. The as-built drawings shall be created using the marked-up prints as reference. A Mylar pencil plot of the floor plans with devices, equipment and notes only will be given to the CCTV Contractor. The CCTV Contractor shall be required to draw in the as-built conduit, wirefill, color coding and miscellaneous changes to create “as built” Mylar drawings from these.

1.7 WARRANTY

A. The CCTV Contractor shall guarantee all materials and workmanship for a period of one year after acceptance of the building by the Owner. The CCTV Contractor shall make all necessary repairs, adjustments, and replacements at no cost to the Owner during this period. Provide a written letter of warranty to Micron on all equipment supplied by the CCTV Contractor outlining start-up dates, duration of warranty and terms of warranty. Refer to Section 01700 for additional requirements.

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

A. The CCTV Contractor shall furnish all equipment required for a complete installation which is not supplied by the Owner or other contractors as part of the requirements listed in other sections of this specification. Where the CCTV Contractor is responsible for installing equipment furnished by the Owner or others, he will take full responsibility for the care and security of that equipment upon receipt of the equipment at the Job Site.

B. All equipment furnished by the CCTV Contractor shall be kept secure and dry at the job site. Equipment that can be affected by ambient conditions shall be stored in a conditioned environment. The CCTV Contractor shall provide all required storage facilities for this equipment on the jobsite.

C. All CCTV Contractor furnished equipment shall be new and left in their original container until installed. The owner reserves the right to inspect this equipment prior to installation and reject any equipment not cared for in an appropriate or defined fashion.

D. Refer to Section 16010 for additional requirements.

2.2 CCTV CAMERAS

A. Cameras :

1). Fixed: Small dome POE. Compatible with Verint VMS 7.5 and capable of at least 106°. Lux 0.25 color and 0.06 BW or IR illumination ≥ 2mp. H.264 compression. Environmental housing as needed

2). PTZ: VMS Compatible as mentioned in fixed category. Capable of 30X optical zoon. Outdoor ready.

a).

2.3 CONDUIT, RACEWAY AND FITTINGS

A. Conduit: All interior conduits shall be EMT, size noted on Drawings as required by NEC cable fill. Exterior conduit shall be rigid steel. Conduit shall meet the following requirements.

2.4 All conduits shall be UL listed with approvedfittings.

A. Conduit shall be sized and supported in accordance with the NEC. Sizes shown on the drawings are to be used for bidding and should conform to the NEC; however, the Contractor shall be responsible for installing the correct size conduit. Tube fill in conduits shall be limited to 40% fill. Minimum size conduit is 3/4 inch,

1). Running threads shall not be permitted and approved threaded couplings shall be used on full weight conduit. Each end of any conduit terminating in a box of any kind shall be provided with a lock-nut inside and outside of the box and with an approved insulating-type bushing.

2). Conduit bends shall be the long radius type without kinks, flattening or crushing. For all bends in conduits 1 inch or larger standard elbows shall be used except where long pulls are involved. Exposed conduits shall be run at right angles or parallel to walls and floors indoors and to the supporting structures outdoors and all changes in direction shallbe made with conduit fittings or standard elbows. Conduit runs outdoors shall be installed with a slight incline in such a manner as to insure against trouble from the collection of trappedcondensation.

3). For above grade conduit installations, as a minimum pull boxes shall be installed in all straight runs of conduit a minimum of every 100 feet; in all runs with one right angle bend a minimum of 75 feet; in all runs with two or three right angle bends a minimum of 50 feet. This requirement does not relieve the Contractor of the responsibility for providing pull points on all long conduit runs as required, so as not to over-stress and/or damage the wire or cable.

4). Open conduit ends, including those terminating in boxes, shall be plugged or capped as soon as installed, and so maintainedduring construction to prevent the entrance of moisture and dirt. All conduits shall be carefully cleaned and dried inside before the installation of wire.

5). All conduits installed outdoors or where subject to water, must enter equipment and junction boxes from the side of the bottom, wheretop entry is a necessity; watertight conduit hubs shall beused.

6). All runs of rigid steel conduit in excess of 100 feet without a right angle pull point. Shall be provided with an expansion joint.

B. Flexible Metal Conduit:

1). Flexible metal conduit shall be installed where called for in the drawings, to provide a flexible connection to equipment requiring position adjustments, or where fished through walls, above ceiling, or within other building cavities as shown on drawings or approved by the owner, flexible metal conduit shall not be used where EMT can be utilized.

2). Flexible metal conduit shall be UL approved with EL approved angle wedge type insulated throat fittings specifically manufactured foruse with flexible metal conduit. Flexible metal conduit shall be ½ inch minimum.

C. Liquid Tight Flexible Metal Conduit

1). Liquid-tight flexible metal conduit where called for shall be fabricated of galvanized steel flexible tubing, with a thermo-plastic jacket extruded over the tubing, Anaconda Seal-tite extra flexible. Liquid tight flexible metal conduit shall be the same size as the conduit system to which it connects unless noted otherwise.

2). Fittings for liquid light flexible metal conduit shall begalvanized malleable iron.

3). Liquid tight flexible metal conduit shall be used for all flexible connections in the basement and penthouse and for all connections to instruments having a NEMA 3, 4, 7 or 9enclosures.

D. Pull Boxes, Junction Boxes and Equipment:

1). Suitable boxes shall be provided where indicated on drawings, called for in the specifications, or as may be required to facilitate the installation and pulling of wires or cables.

2). Boxes shall be sized to accommodate wiring without sharp bends or overcrowding, and shall in all cases comply with NEC requirementsand the cable manufacturer's requirement.

3). All boxes shall be independently and adequately supported and shall not be supported by the conduit system.

2.5 EQUIPMENT MOUNTING

A. Provide all necessary hangers, brackets, steel frames, equipment racks, and pipe stands necessary to properly support all equipment and conduit as required. Where possible, parallel conduits shall be grouped neatly on structural racks and secured in position.

B. Provide channel support to support panels, terminal boxes, and groups of instruments. Equipment shall not be mounted directly on walls or ceilings.

C. All hangers, brackets and channel supports shall be galvanized steel, Uni-strut.

D. All structural steel members or steel frames shall be hot-rolled carbon steel. Shapes shall conform with types already in use in the facility. Steel members shall be welded to existing structures in accordance with the American Welding Society for procedures, appearance and quality ofwelds.

E. All steel shall be galvanized except steel members welded to existing steel which are presently painted and are used solely to support the new equipment or conduit.

F. All fasteners, supports, clamps and anchors shall be of the correct type for the purpose. Fasteners shall be as follows:

1). In hollow or lath construction; toggle or machine bolts, withsuitable bearing plates.

2). For structural iron, machine bolts or approved clamps. Beam clamps can be used on structures free from vibration.

3). For solid masonry, metallic expansion shields and machine screws. In concrete, provide approved inserts.

4). Power driven fasteners and anchors may be used in areas designated by Micron. Where used, it will be the Contractor's responsibility to use the correct anchor and charge for the type of construction encountered.

G. All brackets, hangers, bolts and nuts shall be galvanized.

H. Twenty percent spare space shall be provided in all racks and each spare space shall be not less than 3 inches for future conduit.

2.6 WIRE AND CABLE

A. All conductors shall be new, stranded copper of 98 percent conductivity and marked with identification in accordance with the regulations of Underwriters Laboratories.

1). Cable shall be as follows:

PART 3 EXECUTION

3.1 GENERAL

A. The CCTV system shall include all devices, components, parts, wire, cable, conduit, etc. as specified herein, or in other Sections of the Specification, and connected to perform all functions and operate according to the system description and function contained in Part 1. The CCTV Contractor shall coordinate with other contractors working in the same areas or providing support services (120 volt AC) to his equipment.

B. The CCTV Contractor shall:

1). Furnish and install all CCTV components along with all appurtenant parts, cables, raceway, mounting brackets, and supports required for a complete and operable system unless otherwise covered under another section.

3.2 EQUIPMENT INSTALLATION AND LABELING REQUIREMENTS

A. Equipment shall be installed in accordance with the Drawings and Specifications and in strict accordance with the manufacturer's recommendations. All equipment which must be serviced, operated or maintained shall be located in fully accessible locations. Locations shown on the drawings are diagrammatic but approximate the ideal location for the equipment at the time of design. Changes to these locations shall be coordinated with the Owner. Micron will assign all camera#'s.

B. All electrical equipment must be located so as to provide minimum working space as required by the NEC.

C. The CCTV Contractor shall coordinate with the Architect as to the exactlocation of all equipment and conduit locations that he canutilize.

3.3 DEVICE INSTALLATION AND LABELING REQUIREMENT

A. All equipment and devices shall have nameplates. Nameplates shall be permanently adhered to the device with screws or double sidedtape specifically made for this application. Nameplates shall include the device tag numbers and, if noted on the Drawings, the equipmentdescription.

3.4 CONDUIT/WIRE INSTALLATION AND LABELING

A. All wire and cable shall be continuous from terminal to device with no splices.

B. All cables and wire shall have wire labels at each end and in junction boxes which shall be Brady Data type. Wire, labels shall include tag numbers as listed on the interconnection diagram.

C. All cables within the relay panels should be run in slotted plastic wire duct. Where wire duct is not provided, cables shall be installed using Thomas and Betts (T&B) nylon ty-wraps and anchor blocks.

D. All control wiring shall be verified as to field device and cable integritybefore being terminated at either end per the followingprocedure:

1). Attach DVM across control cable at security panel end.

2). Have person at device end short wires together: verify that continuity is established.

3). Isolate wires at device and check for continuity from wire toshield, shield to ground, and wire to ground.

4). If continuity, is observed on any of the readings in step #3, correct problem or replace cable.

5). If all readings in step #3 show, no continuity, then label and terminate cable at this time.

6). When terminating field devices, strip back shield, cut drain wire, and wrap any exposed shielding with black Scotch 33 vinyl tape. Terminate shield drain wire at control device.

E. Wires shall be installed in strict accordance with the National Electrical Code, and in particular with Article 725. All communication, signal and instrument cable with insulation of less than 600 volts (Class 2 and 3) shall be kept separate by at least 2” from all power and Class 1 control circuits or a partition shall be installed between these classes of cables.

F. Where adjacent terminal blocks are jumpered together as shown on the drawings, standard plug-in jumpers shall be provided and installed by the security contractor. Wire jumpers shall be used between fused terminal blocks and non-adjacent terminal blocks, and shall have one wire label in the middle where 6" or less, and a wire label on both ends where more than 6”.

G. Any penetration of a fire wall shall be sealed to maintain the fire rating of the wall.

H. Pull boxes shall be 4 inches square by 2-1/8 inches deep for all installations except where 1-1/4 inch EMT is used. Where 1-1/4 inch EMT connects to a pull box, the box shall be 4-11/16 inches square by 2-1/8 inches deep. All boxes shall have a sheet metal cover plate and a nameplate identifying the number for cables in the box. Refer to the interconnection diagrams for tag numbers.

I. All 120 volt power cables shall be installed in separate raceway or enclosures from low voltage cables.

END OF SECTION

SECTION 17000
INSTRUMENTATION & CONTROL SYSTEM (ICS) GENERAL INFORMATION

PART 1 - GENERAL

1.1 SUMMARY

A. The following is a detailed description of Engineering services to be provided by the Instrumentation & Controls System Installation Contractor. This term, ICS, I &C may all be used interchangeably. All work is to be performed per drawings and specifications.

B. This Section, 17000 applies to all sections.

1.2 SCOPE

A. I &C contractor shall provide instrumentation, equipment, installation, technical services, as shown on drawings, and/or Instrument Schedules, and /or Specifications. The I &C contractor shall coordinate specifications for the application and submit final instrument data sheets for approval by Micron (MTI) prior to procurement. Procurement of any item without the written approval from MTI is prohibited, and MTI shall have no obligation to pay for any items procured without approval.

B. I &C contractor shall provide and install PLC's and PLC Remote Input / Output (RIO), panels, hardware, field devices. Contractor to provide systems integration unless otherwise noted in design documents.

C. I &C contractor shall provide panels, power supplies, remote emergency shutdown stations, hardware, installation materials, so as to provide a complete Chemical Distribution Control system. MTI shall provide Chemical Distribution system programming.

D. Telecom contractor shall provide (2) CAT 6 cables and pull string in a single conduit for ethernet runs between each PLC and telecom control room. One CAT 6 will be terminated and landed on CPU, the second CAT 6 will be left unterminated as a spare.

E. I &C contractor shall provide panels, power supplies, hardware, installation materials, so as to provide a complete Chemical Distribution Control Tracetek leak detection system. MTI shall provide Chemical Distribution leak detection system programming.

F. Provide for receipt and storage of Instrument Equipment. Receive, maintain inventory, and inspect equipment for damage and shortage. Prior to acceptance, verify equipment conforms to specified requirements. Report damaged equipment to Owner within five days of receipt or assume full responsibility for equipment replacement. Unload equipment using methods, which minimize damage. Store equipment in safe, dry location, protect from the elements, and in accordance with manufacturer recommendations. Deliver manuals, received with instruments to the Owner.

G. Provide calibration and operational hardware bench checks of field instruments. This will prevent installation of defective instruments in the field. Provide certified calibration documentation to the owner.

H. Provide and install conduit, wire, raceways, and accessories to power and interconnect instrumentation and ICS equipment. Note areas marked Class 1, Div. 2.

I. Provide and install pneumatic tubing, fittings, and accessories to power and interconnect pneumatic instrumentation.

J. Provide and install control panels unless otherwise noted in design documents.

K. Provide “red-marked” final P&IDs to reflect “as-built” conditions. P&IDs provided in this package shall be used as a base line. The I &C contractor shall operate in conjunction with MTI personnel to accomplish successful completion of project.

L. Provide field checkout between the Field Termination Assembly Points and field instruments in order to verify that the instrument loop operates correctly (in the case of valves or actuators, verify correct direction per the Final Sequence of Operation).

M. Provide for system startup of each Process, in a coordinated approved manner by MTI.

N. Provide for generation of Final Acceptance Test procedure. Submit to MTI for review/acceptance of format prior to performing test.

O. Provide for performance of Final Acceptance Test, witnessed by MTI for each Process.

1.3 QUALITY ASSURANCE

A. Regulatory Requirements: Conform with requirements of National Electrical Code, National Fire Protection Association, and Local Codes.

B. Standards: Comply with following requirements and recommendations wherever applicable.

1.4 SUBMITTALS

A. Manufacturers Data as required in individual Sections. Submit substitutions for specified instrumentation for approval by M.T.I. prior to procurement.

B. Provide for design and submittal of the following Data Items (five sets of electronic and hard copy required) I&C contractor shall follow naming rules, nomenclature for ICS according to MTI standards, providing the following:

1. Final “red-marked” P&IDs
2. Final Sequence of Operations
3. Final “red-marked” I/O database
4. Data sheets for field instruments (review with Owner prior to procurement)
5. Instrument Loop Wiring Diagrams
6. Instrument Wiring and Tubing Connection Drawings
7. Project Schedule (show P&ID/Sequence of Operation reviews, Simulation Demo Reviews, Data Item Submittal/Review Dates etc., order of process development to coincide with major Construction Milestones etc.)
8. Weekly Progress/Status Reports

1.5 DELIVERY, STORAGE, AND HANDLING OF PREPURCHASED EQUIPMENT

A. Receive, maintain inventory, and inspect equipment for damage and shortage.

B. Prior to acceptance, verify equipment conforms to specified requirements. Report damaged equipment to MTI within five days of receipt or assume full responsibility for equipment replacement.

C. Unload equipment using methods which minimize damage.

D. Store equipment in safe, dry location, protect from the elements, and in accordance with manufacturer recommendations.

E. Deliver manuals, received with instruments to the MTI.

1.6 CONTRACTOR QUALIFICATIONS

A. Demonstrate past semiconductor facility experience and capability to provide a complete and operational system to the satisfaction of MTI .

END OF SECTION 17000